

Pull request summary: beam / rigid-body coupling

1 October 2026 · Colm Mc Alister

beamFoam `colm_6D0F_solver` → `openfoam-v2306` (solids4foam/beamFoam#49). Branch point: bf70a24 2025-04-14 Merge pull request #15 from solids4foam/openfoam-v2306-refactorEvolve. 117 commits plus 12 merges.

moorFV `feature-beamFoamCoupling-colm` → `main` (solids4foam/moorFV#16). Branch point: 1ec54c8 2025-02-25 Merge pull request #9 from solids4foam/branchDev. 14 commits.

What this document is. Every commit since the two branches left their base branches, in chronological order, with:

- a short explanation of what the diff does;
- every file it changes;
- for code and scripts, the line ranges changed.

beamFoam and moorFV commits are interleaved by time and tagged **beamFoam** / **moorFV**. Section 3 summarises the net moorFV changes; section 5 lists open issues found along the way; section 6, at the end, sets out the plan for Phase 5 (several mooring lines).

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1. Overview

Timeline

Period	beamFoam commits	moorFV commits	Authors
2.1 June – September 2025: beamFoam refactoring, function objects, code removal and tutorials	65	0	Amirhossein Taran, Philip Cardiff, Seevani Bali
2.2 November – December 2025: momentum contributions, spring boundary condition, build fixes	24	1	Amir Taran, Colm Mc Alister, Philip Cardiff, Seevani Bali
2.3 February – August 2026: first rigid-body coupling attempt (Colm Mc Alister, with Codex)	17	4	Colm Mc Alister
2.4 30 September – 1 October 2026: review and monolithic coupling, Phases 0–4 (Colm Mc Alister, with Claude)	11	9	Colm Mc Alister

What the branches add, in brief

- **2025, beamFoam (Seevani Bali, Philip Cardiff, Amirhossein Taran).**
 - **New:** cross-section classes for v2306, one `createBeamMesh` utility, four function objects (convergence, displacements, energy, forces/moments), per-beam material properties, and a `beamMomentumContribution`

framework (Morison drag, ground contact).

- **Solver changes:** a linear spring boundary condition with a time-varying offset force, a refactored solver with `d2dt2` scheme names and normalised residuals, and a fix to the Euler inertia terms.
- **Removed:** unused contact, pulley, plasticity, bunching and boundary-condition code.
- **Repository:** OFW-paper tutorials, READMEs, a GitHub Actions build and `Allwmake` fixes.
- **February – August 2026 (Colm, with Codex).**
 - Six rigid-body unknowns appended to the BlockEigen system, and rigid-body step data passed from moorFV.
 - A "staged" and an experimental "fullyImplicit" coupling mode, history files, efficiency plots and interFoam cases.
 - In effect still a partitioned scheme: the body block is an identity, and `FvBeamNewmark` recomputes the body state afterwards.
- **30 September – 1 October 2026 (Colm, with Claude).**
 - A review of the earlier path and a development plan.
 - **Phase 0:** the `rigidBodyEnd` body inside beamFoam, plus two bug fixes.
 - **Phase 1:** monolithic translation in BlockEigen. **Phase 2:** rotation.
 - **Phase 3:** moorFV's `beamFoamCoupled` solver.
 - **Phase 4:** validation and stress tests in interFoam.

How to read the line ranges. Ranges are line numbers in the file *after* that commit (from `git show -U0`). `-n` at `L` means `n` lines were removed just after line `L`, with nothing added there. Generated tutorial files (fields, meshes, time directories, logs) are grouped by case directory with a file count. Tutorial scripts (`.py` , `Allrun` ...) are listed like code. "bin" marks a binary file such as a PDF or image. Path prefixes `src/wireBunchingModels/` (beamFoam) and `src/sDoFRGBFvBeam/` (moorFV) are greyed out.

Net change against the base branches

beamFoam: 2139 files, +131,809,280 / -37,560 lines; code and scripts 246 files, +17,412 / -37,537. Most of the line count is generated tutorial output.

moorFV: 14 files, +697 / -16 lines; code 14 files, +697 / -16.

beamFoam: net changes to code, scripts and documents

File	Lines	Type
.github		
<code>.github/workflows/build.yml</code>	+42 -0	code
(top level)		
<code>Allwmake</code>	+20 -2	code
<code>README.md</code>	+9 -23	doc
applications		
<code>applications/utilities/createBeamMesh/Make/files</code>	+4 -0	code
<code>applications/utilities/{createCircularBeamMesh => createBeamMesh}/Make/options</code>	+4 -0	code
<code>applications/utilities/createBeamMesh/createBeamMesh.C</code>	+592 -0	code
<code>applications/utilities/{createCircularBeamMesh => createBeamMesh}/printMeshSummary.H</code>	+0 -0	code
documentation		
<code>documentation/BEAM_ATTACHMENT_KINEMATICS_OWNERSHIP.md</code>	+104 -0	doc
<code>documentation/BLOCK_EIGEN_RIGID_BODY_COUPLING_PROGRESS.md</code>	+361 -0	doc
<code>documentation/BLOCK_EIGEN_SOLVER_COUPLING_CHANGES.md</code>	+165 -0	doc
<code>documentation/monolithicCouplingPlan.pdf</code>	bin	doc
<code>documentation/outerCorrectorsAndNewmark.pdf</code>	bin	doc
<code>documentation/rigidBodyAndBeamCoupledSolverDevelopmentLog.txt</code>	+541 -0	doc
<code>documentation/rigidBodyCouplingPRSummary.pdf</code>	bin	doc

documentation/rigidBodyImplicitAttachmentCoefficientChanges.txt	+181 -0	doc
documentation/rigidBodyKinematicCouplingDevelopment.txt	+217 -0	doc
documentation/source/README.md	+15 -0	doc
documentation/source/figs.py	+36 -0	doc
documentation/source/growth.png	bin	doc
documentation/source/model.py	+43 -0	doc
documentation/source/monolithicCouplingPlan.html	+319 -0	doc
documentation/source/newton.png	bin	doc
documentation/source/order.py	+13 -0	doc
documentation/source/outerCorrectorsAndNewmark.html	+243 -0	doc
src/wireBunchingModels/Make		
src/wireBunchingModels/Make/files	+18 -11	code
src/wireBunchingModels/Make/options	+1 -0	code
src/wireBunchingModels/beamModels		
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+213 -124	code
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+136 -76	code
src/wireBunchingModels/beamModels/beamModel/crossSections/annularSector/annularSector.C	+352 -353	code
src/wireBunchingModels/beamModels/beamModel/crossSections/annularSector/annularSector.H	+43 -43	code
src/wireBunchingModels/beamModels/beamModel/crossSections/circle/circle.C	+17 -17	code
src/wireBunchingModels/beamModels/beamModel/crossSections/circle/circle.H	+80 -80	code
src/wireBunchingModels/beamModels/beamModel/crossSections/crossSection/crossSectionModel.C	+8 -1	code
src/wireBunchingModels/beamModels/beamModel/crossSections/crossSection/crossSectionModel.H	+36 -36	code
src/wireBunchingModels/beamModels/beamModel/crossSections/crossSection/newCrossSectionModel.C	+3 -4	code
src/wireBunchingModels/beamModels/beamModel/crossSections/polygon/polygon.C	+12 -12	code
src/wireBunchingModels/beamModels/beamModel/crossSections/polyline/polyline.C	+6 -6	code
src/wireBunchingModels/beamModels/beamModel/crossSections/rectangle/rectangle.C	+315 -306	code
src/wireBunchingModels/beamModels/beamModel/crossSections/rectangle/rectangle.H	+64 -50	code
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/RigidBodyCouplingData.H	+246 -0	code
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+198 -128	code
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+215 -43	code
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamBoundaryConditions.C	+431 -469	code
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+1018 -567	code
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C	+691 -0	code
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.C	+592 -0	code
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.H	+450 -0	code
src/wireBunchingModels/beamModels/fvPatchFields/{myFixedRotation/myFixedRotationFvPatchVectorField.C => fixedRotation/fixedRotationFvPatchVectorField.C}	+21 -21	code

src/wireBunchingModels/beamModels/fvPatchFields/{myFixedRotation/myFixedRotationFvPatchVectorField.H => fixedRotation/fixedRotationFvPatchVectorField.H}	+17 -17	code
src/wireBunchingModels/beamModels/fvPatchFields/followerForceBeamDisplacementNR/followerForceBeamDisplacementNRFvPatchVectorField.C	+11 -11	code
src/wireBunchingModels/beamModels/fvPatchFields/followerForceBeamDisplacementNR/followerForceBeamDisplacementNRFvPatchVectorField.H	+3 -0	code
src/wireBunchingModels/beamModels/fvPatchFields/linearSpringForceBeamDisplacementNR/linearSpringForceBeamDisplacementNRFvPatchVectorField.C	+330 -0	code
src/wireBunchingModels/beamModels/fvPatchFields/{myFollowerForceBeamDisplacementNR/myFollowerForceBeamDisplacementNRFvPatchVectorField.H => linearSpringForceBeamDisplacementNR/linearSpringForceBeamDisplacementNRFvPatchVectorField.H}	+42 -40	code
src/wireBunchingModels/beamMomentumContribution		
src/wireBunchingModels/{beamContactModel/contactModels/normalContactModels/normalContactModel/newNormalContactModel.C => beamMomentumContribution/beamMomentumContribution.C}	+46 -23	code
src/wireBunchingModels/beamMomentumContribution/beamMomentumContribution.H	+169 -0	code
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.C	+260 -0	code
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.H	+160 -0	code
src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.C	+219 -0	code
src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.H	+153 -0	code
src/wireBunchingModels/functionObjects		
src/wireBunchingModels/functionObjects/beamConvergenceData/beamConvergenceData.C	+175 -0	code
src/wireBunchingModels/functionObjects/beamConvergenceData/beamConvergenceData.H	+134 -0	code
src/wireBunchingModels/functionObjects/beamDisplacements/beamDisplacements.C	+214 -0	code
src/wireBunchingModels/functionObjects/beamDisplacements/beamDisplacements.H	+142 -0	code
src/wireBunchingModels/functionObjects/beamEnergyData/beamEnergyData.C	+277 -0	code
src/wireBunchingModels/functionObjects/beamEnergyData/beamEnergyData.H	+133 -0	code
src/wireBunchingModels/functionObjects/beamForcesMoments/beamForcesMoments.C	+221 -0	code
src/wireBunchingModels/functionObjects/beamForcesMoments/beamForcesMoments.H	+142 -0	code
src/wireBunchingModels/numerics		
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+805 -157	code
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+77 -12	code
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/original_BlockEigenSolverOF.C	+576 -0	code
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/original_BlockEigenSolverOF.H	+152 -0	code
tutorials		
tutorials/3DdynamicCantilever/Allclean	+12 -0	script
tutorials/3DdynamicCantilever/Allrun	+24 -0	script
tutorials/3DdynamicCantilever/README.md	+81 -0	doc

tutorials/3DdynamicCantilever/allPlots.gnuplot	+48 -0	script
tutorials/3DdynamicCantilever/cantilever_abaqus_50percent.txt	+1002 -0	doc
tutorials/3DdynamicCantilever/dispPlotDT.gnuplot	+32 -0	script
tutorials/3DdynamicCantilever/dispPlotMesh.gnuplot	+30 -0	script
tutorials/3DdynamicCantilever/runSweep.sh	+65 -0	script
tutorials/3DdynamicCantilever/runTimeSweep.sh	+71 -0	script
tutorials/beamAndSpringInSeries/Allclean	+10 -0	script
tutorials/beamAndSpringInSeries/Allrun	+10 -0	script
tutorials/cantilever/Allclean	+10 -0	script
tutorials/cantilever/Allrun	+14 -0	script
tutorials/cantilever/README.md	+63 -0	doc
tutorials/cantilever/allPlots.gnuplot	+25 -0	script
tutorials/cantilever/displacementPlot.pdf	bin	doc
tutorials/cantileverFollowerForce/Allclean	+12 -0	script
tutorials/cantileverFollowerForce/Allrun	+18 -0	script
tutorials/cantileverFollowerForce/README.md	+56 -0	doc
tutorials/cantileverFollowerForce/allPlots.gnuplot	+28 -0	script
tutorials/cantileverFollowerForce/dispPlot.gnuplot	+40 -0	script
tutorials/cantileverFollowerForce/runSweep.sh	+69 -0	script
tutorials/cantileverSpring/Allclean	+10 -0	script
tutorials/cantileverSpring/Allrun	+10 -0	script
tutorials/freeFlexibleBeam/Allclean	+12 -0	script
tutorials/freeFlexibleBeam/Allrun	+21 -0	script
tutorials/freeFlexibleBeam/README.md	+62 -0	doc
tutorials/freeFlexibleBeam/allPlots.gnuplot	+51 -0	script
tutorials/plot_rigidBodyAndBeam_algorithm_efficiency_new_vs_old.py	+943 -0	script
tutorials/plot_rigidBodyAndBeam_new_vs_old.py	+341 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam/Alltest	+3 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam/check_block_eigen_authoritative_regression.py	+196 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_ForceCoupledSolver/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_ForceCoupledSolver/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_ForceCoupledSolver/Alltest	+3 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_ForceCoupledSolver/check_block_eigen_authoritative_regression.py	+196 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_ForceCoupledSolver/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_compareSixDoF_Eigen/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_compareSixDoF_Eigen/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_compareSixDoF_Eigen/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_fullTime/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_fullTime/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_fullTime/Alltest	+3 -0	script

tutorials/rigidBodyAndBeam/rigidBodyAndBeam_fullTime/check_block_eigen_authoritative_regression.py	+196 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_fullTime/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_short/Allclean	+16 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_short/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_shortTime/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_shortTime/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_shortTime/Alltest	+3 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_shortTime/check_block_eigen_authoritative_regression.py	+196 -0	script
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_shortTime/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled/Alltest	+3 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled/checkCoupledState.py	+108 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/Alltest	+3 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/checkCoupledState.py	+108 -0	script
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam_loop/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam_loop/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam_loop/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam_loop_nOuter8/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam_loop_nOuter8/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam_loop_nOuter8/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam_monolithic/Allclean	+19 -0	script
tutorials/rigidBodyAndBeam_monolithic/Allrun	+10 -0	script
tutorials/rigidBodyAndBeam_monolithic/Alltest	+3 -0	script
tutorials/rigidBodyAndBeam_monolithic/check_block_eigen_authoritative_regression.py	+196 -0	script
tutorials/rigidBodyAndBeam_monolithic/plot_rigid_body_displacement_comparison.py	+279 -0	script
tutorials/rigidBodyAndBeam_stress/Allclean	+4 -0	script
tutorials/rigidBodyAndBeam_stress/Allrun	+38 -0	script
tutorials/rigidBodyAndBeam_stress/makeCases	+121 -0	script
tutorials/rigidBodyAndBeam_stress/summarise.py	+149 -0	script
tutorials/rigidBodyEnd/Allclean	+9 -0	script
tutorials/rigidBodyEnd/Allrun	+26 -0	script
tutorials/rigidBodyEnd/README.md	+64 -0	doc
tutorials/rigidBodyEnd/axialOscillation/Allclean	+7 -0	script
tutorials/rigidBodyEnd/axialOscillation/Allrun	+6 -0	script
tutorials/rigidBodyEnd/checkRigidBodyEnd.py	+472 -0	script
tutorials/rigidBodyEnd/compoundPendulum/Allclean	+7 -0	script
tutorials/rigidBodyEnd/compoundPendulum/Allrun	+6 -0	script

tutorials/rigidBodyEnd/freeRotation/Allclean	+7 -0	script
tutorials/rigidBodyEnd/freeRotation/Allrun	+6 -0	script
tutorials/rigidBodyEnd/hangingBody/Allclean	+7 -0	script
tutorials/rigidBodyEnd/hangingBody/Allrun	+6 -0	script
tutorials/rigidBodyEnd/hangingMass/Allclean	+7 -0	script
tutorials/rigidBodyEnd/hangingMass/Allrun	+6 -0	script
tutorials/rigidBodyEnd/pendulum/Allclean	+7 -0	script
tutorials/rigidBodyEnd/pendulum/Allrun	+6 -0	script
Deleted files, grouped by directory		
applications/utilities/createCircularBeamMesh/	-605	20 files
src/wireBunchingModels/beamContactModel/	-3650	4 files
src/wireBunchingModels/beamContactModel/allAngleContact/	-2883	7 files
src/wireBunchingModels/beamContactModel/conicalPulleyContact/	-4283	3 files
src/wireBunchingModels/beamContactModel/contactModels/	-3237	22 files
src/wireBunchingModels/beamContactModel/lineContact/	-2803	7 files
src/wireBunchingModels/beamContactModel/pointContact/	-1154	2 files
src/wireBunchingModels/beamContactModel/toroidalPulleyContact/	-769	2 files
src/wireBunchingModels/beamModels/beamModel/	-1573	4 files
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/	-5443	5 files
src/wireBunchingModels/beamModels/fvPatchFields/	-4212	18 files
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/	-3329	8 files
src/wireBunchingModels/beamsBunchingModel/	-987	2 files

beamFoam: net changes to tutorial cases

Generated and setup files per case, from the branch point to now. Scripts are listed in the table above.

Tutorial case	Lines	Files
tutorials/3DdynamicCantilever/	+673 -0	11
tutorials/beamAndSpringInSeries/	+522 -0	13
tutorials/cantilever/	+463 -0	12
tutorials/cantileverFollowerForce/	+1394 -0	11
tutorials/cantileverSpring/	+486 -0	11
tutorials/freeFlexibleBeam/	+443 -0	12
tutorials/rigidBodyAndBeam/rigidBodyAndBeam/	+946629 -0	97
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_ForceCoupledSolver/	+946627 -0	97
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_compareSixDoF_Eigen/	+946623 -0	97
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_coupledSolver/	+121367244 -0	670
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_fullTime/	+946629 -0	97
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_short/	+948569 -0	96
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_shortTime/	+946629 -0	97
tutorials/rigidBodyAndBeam_beamFoamCoupled/	+946628 -0	97
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/	+946634 -0	97
tutorials/rigidBodyAndBeam_loop/	+946625 -0	97
tutorials/rigidBodyAndBeam_loop_nOuter8/	+946631 -0	97
tutorials/rigidBodyAndBeam_monolithic/	+946630 -0	97
tutorials/rigidBodyAndBeam_stress/	+7 -0	1
tutorials/rigidBodyEnd/	+1 -0	1
tutorials/rigidBodyEnd/axialOscillation/	+365 -0	10
tutorials/rigidBodyEnd/compoundPendulum/	+372 -0	10

tutorials/rigidBodyEnd/freeRotation/	+369 -0	10
tutorials/rigidBodyEnd/hangingBody/	+374 -0	10
tutorials/rigidBodyEnd/hangingMass/	+361 -0	10
tutorials/rigidBodyEnd/pendulum/	+365 -0	10

2. Chronological commit log

2.1 June – September 2025: beamFoam refactoring, function objects, code removal and tutorials

Work by Seevani Bali, with Philip Cardiff and Amirhossein Taran, on the `openfoam-v2306-seevani` line. It is merged into this branch, but not yet into `openfoam-v2306` , so it shows up in the PR diff.

beamFoam

c4ff656

2025-06-06

Seevani Bali

adding crossSections class compatible of 2306 version, modify beamModel.C/.H by uncommmenting crossSection class functions + createMultipleBeamMesh utility

Re-enables the crossSectionModel family (circle, rectangle, polygon, polyline, annularSector) for OpenFOAM v2306+ and makes beamModel build per-beam cross-sections from a 'beams' list. Also adds a createMultipleBeamMesh utility.

- Make/files: compiles crossSectionModel, circle, rectangle, quadrilateral, polygon, annularSector, polyline
- beamModel.C/.H: crossSections_ PtrList and crossSections() accessor un-commented; R and A() now come from each beam's crossSectionModel
- crossSection sources: run-time selection table and tmp/.ref() usage updated to the newer OpenFOAM API
- createMultipleBeamMesh.C: new utility that meshes several beams, one per 'beams' entry

File	Lines	Changed line ranges (after the commit)
applications/utilities/createMultipleBeamMesh/Make/files	+4 -0	1–4
applications/utilities/createMultipleBeamMesh/Make/options	+13 -0	1–13
applications/utilities/createMultipleBeamMesh/createMultipleBeamMesh.C	+587 -0	1–587
src/wireBunchingModels/Make/files	+8 -1	3–10
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+77 -77	243, 363–364, -1 at 366, 369–372, 374, 376–379, 381–383, 391–393, 418–448, 478 (+20 more)
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+46 -38	-1 at 42, 52, 122, 395–398, 414–421, -1 at 423, 428–435, -1 at 437, 442–449, -1 at 451 (+4 more)
src/wireBunchingModels/beamModels/beamModel/crossSections/annularSector/annularSector.C	+86 -86	54, 56, 60, 68, 70, 72, 74–75, 83, 85–94, 142 (+42 more)
src/wireBunchingModels/beamModels/beamModel/crossSections/circle/circle.H	+3 -3	167–169
src/wireBunchingModels/beamModels/beamModel/crossSections/crossSection/crossSectionModel.C	+8 -1	102–109
src/wireBunchingModels/beamModels/beamModel/crossSections/crossSection/newCrossSectionModel.C	+3 -4	49, 51, 69
src/wireBunchingModels/beamModels/beamModel/crossSections/polygon/polygon.C	+12 -12	425, 439, 448, 457, 466, 475, 516, 667–669, 671–672
src/wireBunchingModels/beamModels/beamModel/crossSections/polyline/polyline.C	+6 -6	183, 334–336, 338–339
src/wireBunchingModels/beamModels/beamModel/crossSections/rectangle/rectangle.C	+6 -6	156, 377–379, 381–382
src/wireBunchingModels/beamModels/beamModel/crossSections/rectangle/rectangle.H	+10 -4	130–135, 178–181

beamFoam 6e5bbc4 2025-06-06 Seevani Bali

Renamed Lambda to Lambdaf in followerBeam BC to make it consistent with previous coding style changes to Lambda notation

Renames the face rotation-tensor variable in the follower-force boundary condition from Lambda to Lambdaf and looks up the 'Lambdaf' surface field, matching the solver's naming.

- followerForceBeamDisplacementNRFvPatchVectorField.C: lookupPatchField<surfaceTensorField>("Lambdaf") replaces "Lambda"

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/fvPatchFields/followerForceBeamDisplacementNR/followerForceBeamDisplacementNRFvPatchVectorField.C	+11 -11	211, 213, 221–223, 226–228, 231–233

beamFoam baebd59 2025-06-06 Seevani Bali

Adding followerBeam test case for ofw-paper

Adds a cantilever follower-end-force test case under tutorials/ofw-paper (fields, beamProperties, load tables, Allrun/Allclean).

File	Lines	Changed line ranges (after the commit)
tutorials/ofw-paper/myCantilever_followerEndForce/Allclean	+8 -0	1–8
tutorials/ofw-paper/myCantilever_followerEndForce/Allrun	+15 -0	1–15
tutorials/ofw-paper/myCantilever_followerEndForce/ (case files)	+468 -0	12 files

beamFoam 075bae9 2025-07-23 Seevani Bali

Correcting indentation in followerForce fvPatchField

Whitespace-only fix (a space before '=') in the follower-force boundary condition.

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/fvPatchFields/followerForceBeamDisplacementNR/followerForceBeamDisplacementNRFvPatchVectorField.C	+1 -1	211

beamFoam d7fe04b 2025-07-24 Seevani Bali

Adjusting indentation to crossSections/annularSector

Formatting-only change: re-indents code and commented-out lines in annularSector.C; behaviour is unchanged.

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/crossSections/annularSector/annularSector.C	+19 -20	71, 75–83, 87–93, 95, 154, -1 at 157

Adding functionObject

Adds the beamDisplacements function object, which writes the average W displacement (x, y, z) on a named patch to a time-history file.

- functionObjects/beamDisplacements: new class; reads 'historyPatch' and writes Time, w_x, w_y, w_z
- Ships its own Make/files and Make/options

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/functionObjects/beamDisplacements/Make/files	+3 -0	1-3
src/wireBunchingModels/functionObjects/beamDisplacements/Make/options	+7 -0	1-7
src/wireBunchingModels/functionObjects/beamDisplacements/beamDisplacements.C	+212 -0	1-212
src/wireBunchingModels/functionObjects/beamDisplacements/beamDisplacements.H	+142 -0	1-142

Adding functionObject beamForcesMoments

Adds the beamForcesMoments function object, which writes the average force Q and moment M on a named patch to a time-history file.

- functionObjects/beamForcesMoments: new class; reads 'historyPatch' and writes Q_x..Q_z and M_x..M_z

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/functionObjects/beamForcesMoments/Make/files	+3 -0	1-3
src/wireBunchingModels/functionObjects/beamForcesMoments/Make/options	+7 -0	1-7
src/wireBunchingModels/functionObjects/beamForcesMoments/beamForcesMoments.C	+221 -0	1-221
src/wireBunchingModels/functionObjects/beamForcesMoments/beamForcesMoments.H	+142 -0	1-142

Adding functionObject beamEnergyData

Adds the beamEnergyData function object, which writes the beam's internal energy and its linear and angular kinetic energy each time step.

- functionObjects/beamEnergyData: computes energies from Gamma, K, CQ, CM, U and Omega; writes beamEnergyData.dat

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/functionObjects/beamEnergyData/Make/files	+3 -0	1-3
src/wireBunchingModels/functionObjects/beamEnergyData/Make/options	+7 -0	1-7
src/wireBunchingModels/functionObjects/beamEnergyData/beamEnergyData.C	+277 -0	1-277
src/wireBunchingModels/functionObjects/beamEnergyData/beamEnergyData.H	+133 -0	1-133

beamFoam 3273c83 2025-07-24 Seevani Bali

Adding functionObject beamConvergenceData

Adds the beamConvergenceData function object, which writes the number of Newton-Raphson corrections used in each time step.

- functionObjects/beamConvergenceData: writes Time and nCorrectors to beamConvergenceData.dat

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/functionObjects/beamConvergenceData/Make/files	+3 -0	1-3
src/wireBunchingModels/functionObjects/beamConvergenceData/Make/options	+7 -0	1-7
src/wireBunchingModels/functionObjects/beamConvergenceData/beamConvergenceData.C	+175 -0	1-175
src/wireBunchingModels/functionObjects/beamConvergenceData/beamConvergenceData.H	+134 -0	1-134

beamFoam 139421a 2025-07-24 Seevani Bali

Update Make folder in src to include functionObjects

Adds the four new beam function objects to the main library build so they compile into the wireBunchingModels library.

- src/wireBunchingModels/Make/files: adds beamEnergyData, beamConvergenceData, beamDisplacements, beamForcesMoments

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+4 -0	32-35

beamFoam 3056142 2025-08-07 Seevani Bali

Remove individual Make folders in functionObjects + added magnitude of displacements in beamDisplacement function object

Deletes the per-function-object Make folders, which are now redundant after 139421a, and adds a displacement-magnitude column to beamDisplacements output.

- beamDisplacements.C: writes mag(avgW) with header 'mag(w)'
- Removes Make/files and Make/options under all four functionObjects directories

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/functionObjects/beamConvergenceData/Make/files	+0 -3	-3 at 0
src/wireBunchingModels/functionObjects/beamConvergenceData/Make/options	+0 -7	-7 at 0
src/wireBunchingModels/functionObjects/beamDisplacements/Make/files	+0 -3	-3 at 0
src/wireBunchingModels/functionObjects/beamDisplacements/Make/options	+0 -7	-7 at 0
src/wireBunchingModels/functionObjects/beamDisplacements/beamDisplacements.C	+4 -2	74-75, 180-181
src/wireBunchingModels/functionObjects/beamEnergyData/Make/files	+0 -3	-3 at 0
src/wireBunchingModels/functionObjects/beamEnergyData/Make/options	+0 -7	-7 at 0
src/wireBunchingModels/functionObjects/beamForcesMoments/Make/files	+0 -3	-3 at 0
src/wireBunchingModels/functionObjects/beamForcesMoments/Make/options	+0 -7	-7 at 0

beamFoam f3fd8ca 2025-08-07 Seevani Bali

Commented unwanted virtual member functions in crossSections/annularSector

Comments out annularSector's plasticity and stress-resultant virtual functions. It also drops polygon and polyline from the library build, which the commit message does not mention.

- annularSector.C/.H: CNM, greenLagrangianStrain, resultantForce/Moment, resultantTangentMatrices commented out
- Make/files: polygon.C and polyline.C no longer compiled

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+0 -3	-1 at 7, -2 at 8
src/wireBunchingModels/beamModels/beamModel/crossSections/annularSector/annularSector.C	+286 -286	166, 175, 225, 236, 249, 257-537
src/wireBunchingModels/beamModels/beamModel/crossSections/annularSector/annularSector.H	+43 -43	174-179, 196-224, 226-233

beamFoam e6d8425 2025-08-07 Seevani Bali

Commented pure virtual functions from abstract class crossSections

Comments out the matching pure virtual functions in the crossSectionModel base class, so derived cross-sections no longer have to implement them.

- crossSectionModel.H: CNM, greenLagrangianStrain, resultantTotalForce, resultantForce, resultantMoment, resultantTangentMatrices commented out

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/crossSections/crossSection/crossSectionModel.H	+36 -36	169-171, 182-188, 190-195, 197-198, 200-209, 211-218

beamFoam ab4ee5a 2025-08-07 Seevani Bali

Commented unwanted virtual member functions in circle and rectangle crossSections classes

Comments out the same unused virtual functions (plus writeVTK) in the circle and rectangle cross-sections.

- circle.C/.H, rectangle.C/.H: plasticity, stress-resultant and writeVTK members commented out

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/crossSections/circle/circle.C	+17 -17	70, 72, 74-75, 82-94, -4 at 95
src/wireBunchingModels/beamModels/beamModel/crossSections/circle/circle.H	+79 -79	64, 67-74, 78, 115, 121, 127, 131, 133, 137, 139 (+6 more)
src/wireBunchingModels/beamModels/beamModel/crossSections/rectangle/rectangle.C	+315 -306	83-91, 136-191, 193-194, 196, 198-199, 201-202, 204-211, 213-238, 240, 242-244 (+31 more)
src/wireBunchingModels/beamModels/beamModel/crossSections/rectangle/rectangle.H	+58 -50	79-86, 147, 153, 159, 168-173, 186-189, 207-235, 237-244

beamFoam 4a6fbcc 2025-08-07 Seevani Bali

Adding one combined mesh utility called createBeamMesh with functionalities both from createCircularBeamMesh and createMultipleBeamMesh

Adds createBeamMesh, a single mesh utility that merges createCircularBeamMesh and createMultipleBeamMesh. It is mostly a reformatted copy of createMultipleBeamMesh, but places the beam axis along x instead of z.

- createBeamMesh.C: reads the 'beams' list with crossSectionModel, length and nSegments per beam
- Beam points are laid out along x; the decomposePatches option is disabled

File	Lines	Changed line ranges (after the commit)
applications/utilities/createBeamMesh/Make/files	+4 -0	1-4
applications/utilities/createBeamMesh/Make/options	+13 -0	1-13
applications/utilities/createBeamMesh/createBeamMesh.C	+592 -0	1-592

beamFoam 71644d5 2025-08-07 Seevani Bali

Removing objective Interpolation keyword

Disables the 'objectiveInterpolation' option, an alternative rotation-update path the code marked as not working, so only the standard update remains. Also expands the reference-configuration error message to point users to createBeamMesh.

- beamModel.C/.H: objectiveInterpolation_ member, keyword and accessor commented out; kCl() accessor tidied
- coupledTotalLagNewtonRaphsonBeamEvolve.C: objective-interpolation branch commented out
- coupledTotalLagNewtonRaphsonBeam.C: error now warns that the deprecated mesh utilities plus transformPoints are no longer supported

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+4 -4	307–310
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+16 -10	32, 184–188, 801–809, 811, -5 at 812
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+9 -2	742–743, 747–753
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+2 -2	260–261
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+30 -30	665–693, 754

beamFoam 33e3e99 2025-08-07 Seevani Bali

Removing objective Interpolation keyword

Cherry-pick of 71644d5 with an identical patch (same patch-id): disables objectiveInterpolation.

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+4 -4	307–310
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+16 -10	32, 184–188, 801–809, 811, -5 at 812
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+9 -2	742–743, 747–753
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+2 -2	260–261
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+30 -30	665–693, 754

beamFoam 28a7c84 2025-08-10 Seevani Bali

Changed deprecated set() to reset() in beamModel.C + added d2dt2Scheme to tutorial case

Replaces deprecated autoPtr::set() with reset() for the gravity pointer, and adds a d2dt2Schemes block to the ofw-paper follower-force tutorial.

- beamModel.C: gPtr_.set -> gPtr_.reset
- tutorials/ofw-paper/.../fvSchemes: d2dt2Schemes default steadyState

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+3 -3	348, 398, 457
tutorials/ofw-paper/myCantilever_followerEndForce/ (case files)	+5 -0	1 file

beamFoam e84b895 2025-08-10 Seevani Bali

Changed deprecated set() to reset() in beamModel.C + added d2dt2Scheme to tutorial case

Cherry-pick of 28a7c84 with an identical patch (same patch-id): set() to reset(), and d2dt2Schemes added to the tutorial.

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+3 -3	348, 398, 457
tutorials/ofw-paper/myCantilever_followerEndForce/ (case files)	+5 -0	1 file

beamFoam 7100bc8 2025-08-10 Seevani Bali

Refactoring of CTLNRB.C/.H/Evolve.C code, adding d2dt2SchemeNames, refactor Newmark and Euler section of code to make it less clumsy and \OFconsistent

Selects time integration from fvSchemes instead of a beamProperties 'newmark' flag, and rewrites the Newmark/Euler inertia terms in the solver.

- coupledTotalLagNewtonRaphsonBeam.C/.H: new ddtSchemeName_ and d2dt2SchemeName_; d2dt2 must be steadyState, Euler or Newmark, with ddt/d2dt2 pairs checked for consistency
- newmark_ flag removed; info message prints the Newmark beta and gamma values
- Evolve.C: Newmark predictor for AccI/U and dotOmega/Omega; scheme-specific inertia source and diagonal coefficients

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+84 -3	634-651, 682-743, 746, 748, 750, 815
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+15 -6	205, 208, 210-219, 222, 229-230, -1 at 343
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+310 -298	174, 180-183, 185-201, 332-351, 353-367, 369-378, 380-457, 459-475, -1 at 476, 478-482 (+31 more)

beamFoam 5cdb1f0 2025-08-10 Seevani Bali

Refactoring of CTLNRB.C/.H/Evolve.C code, adding d2dt2SchemeNames, refactor Newmark and Euler section of code to make it less clumsy and \OFconsistent

Cherry-pick of 7100bc8 with an identical patch (same patch-id): fvSchemes-driven time integration and Newmark/Euler refactor.

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+84 -3	634-651, 682-743, 746, 748, 750, 815
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+15 -6	205, 208, 210-219, 222, 229-230, -1 at 343
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+310 -298	174, 180-183, 185-201, 332-351, 353-367, 369-378, 380-457, 459-475, -1 at 476, 478-482 (+31 more)

beamFoam 4f4a3e6 2025-08-12 Seevani Bali

Change: consistent time schemes flag for Newmark + drag active warning to fatal error + removed submerged fluid loop

Makes Newmark configurable entirely from fvSchemes: it now requires (Newmark, Newmark) for ddt/d2dt2, with beta and gamma read there. Drag on a steady case becomes a fatal error, and the gravity/buoyancy source loop is commented out.

- coupledTotalLagNewtonRaphsonBeam.C: scheme lookups use ddt(W)/d2dt2(W); newmarkBeta(W)/newmarkGamma(W) read from d2dt2Schemes (defaults 0.25/0.5)
- Allowed scheme pairs: (Euler, Euler), (steadyState, steadyState), (Newmark, Newmark)
- Evolve.C: dragActive with a steady scheme now aborts; the (rho - rhoFluid)*g source loop is commented out

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+17 -18	636, 638, 644, 646, 651-652, 702-703, 714, 720, 723-726, 731-733
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+24 -22	295-311, 541, 552, 556-560

beamFoam f9d21a6 2025-08-22 Philip Cardiff

beamModel: remove unused functions

Removes unused increment accessors from beamModel and the solver.

- beamModel.H: pure virtuals currentDisplacementIncrement() and currentRotationIncrement() removed, plus the commented objectiveInterpolation accessor
- coupledTotalLagNewtonRaphsonBeam.C/.H: implementations of both functions deleted

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+0 -12	-6 at 697, -6 at 794
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+1 -56	-55 at 2034, 2037
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+0 -6	-6 at 438

beamFoam c94ad56 2025-08-26 Seevani Bali

removed follwer test case from ofw-paper because I want to rearrange it to tutorials folder directly

Deletes the tutorials/ofw-paper follower-force case so it can be reorganised under tutorials/.

File	Lines	Changed line ranges (after the commit)
tutorials/ofw-paper/myCantilever_followerEndForce/Allclean	+0 -8	-8 at 0
tutorials/ofw-paper/myCantilever_followerEndForce/Allrun	+0 -15	-15 at 0
tutorials/ofw-paper/myCantilever_followerEndForce/ (case files)	+0 -473	12 files

beamFoam f534931 2025-08-26 Seevani Bali

Adding feature that every beam can have local mechanical properties - rho set to default zero - Have to fix this

Allows E, G and rho to be set per beam inside each 'beams' entry, or globally, but not both. The 'beams' list becomes mandatory, rho defaults to zero, and a missing-E/G error does not abort.

- beamModel.H: E_, G_, rho_ become scalarLists; E(bl), G(bl), rho(bl) accessors feed EA, EI, GJ and the inertia terms
- beamModel.C: fatal error if both local and global values are given
- Old nBeams/R/U input paths and U_ commented out
- Missing-E/G FatalErrorInFunction ends with endl rather than abort, so it does not stop the run

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+158 -67	-1 at 143, 244-248, 251-255, 333-336, 417-419, 421, 423, 425, 427, 429-440 (+13 more)
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+41 -25	129, 132-133, 136-137, 140-142, 192, 415-419, 491, 493, 495, 500 (+14 more)

beamFoam a1e5b97 2025-08-26 Seevani Bali

Minor change to fatal error comment

Fixes spacing, line breaks and wording in the local-versus-global material-property error message.

- beamModel.C: message text only

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+4 -4	525-526, 529, 532

Adding Fatal Error check if density is not specified for transient case

Aborts a transient (Euler or Newmark) run if any beam's density is zero.

- coupledTotalLagNewtonRaphsonBeamEvolve.C: loops over beams and raises FatalErrorInFunction when $\rho(b) == 0$

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+15 -0	353-367

Removed the base class beamContactModel and all contact related codes

Deletes the beamContactModel framework and all contact code: line, point, all-angle and pulley contact, plus the normal and friction contact models. It also deletes the solver's applyContacts sources and stray editor and backup files.

- src/wireBunchingModels/beamContactModel/: entire tree removed (about 24k lines)
- coupledTotalLagNewtonRaphsonBeam: applyContacts.C, applyContacts.org_1 and applyLagrangeMultiplierConicalPulleyContact.H removed

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamContactModel/allAngleContact/.ccls-cache/@home@seevani@Desktop@bekaert-bunching-modification-linearisation-partII@src@wireBunchingModels@beamContactModel@lineContact/lineContact.C	+0 -910	-910 at 0
src/wireBunchingModels/beamContactModel/allAngleContact/.ccls-cache/@home@seevani@Desktop@bekaert-bunching-modification-linearisation-partII@src@wireBunchingModels@beamContactModel@lineContact/lineContact.C.blob	bin	binary
src/wireBunchingModels/beamContactModel/allAngleContact/.ccls-cache/@home@seevani@Desktop@bekaert-bunching-modification-linearisation-partII@src@wireBunchingModels@beamContactModel@lineContact/lineContact.H	+0 -433	-433 at 0
src/wireBunchingModels/beamContactModel/allAngleContact/.ccls-cache/@home@seevani@Desktop@bekaert-bunching-modification-linearisation-partII@src@wireBunchingModels@beamContactModel@lineContact/lineContact.H.blob	bin	binary
src/wireBunchingModels/beamContactModel/allAngleContact/.vscode/settings.json	+0 -5	-5 at 0
src/wireBunchingModels/beamContactModel/allAngleContact/allAngleContact.C	+0 -1062	-1062 at 0
src/wireBunchingModels/beamContactModel/allAngleContact/allAngleContact.H	+0 -473	-473 at 0
src/wireBunchingModels/beamContactModel/beamContactModel.C	+0 -2115	-2115 at 0
src/wireBunchingModels/beamContactModel/beamContactModel.H	+0 -447	-447 at 0
src/wireBunchingModels/beamContactModel/beamContactModelConicalPully.C	+0 -652	-652 at 0
src/wireBunchingModels/beamContactModel/beamContactModelTorooidalPully.C	+0 -436	-436 at 0
src/wireBunchingModels/beamContactModel/conicalPulleyContact/conicalPulleyContact.C	+0 -1864	-1864 at 0
src/wireBunchingModels/beamContactModel/conicalPulleyContact/conicalPulleyContact.C.old	+0 -1816	-1816 at 0
src/wireBunchingModels/beamContactModel/conicalPulleyContact/conicalPulleyContact.H	+0 -603	-603 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/backup.tar.xz	bin	binary
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/frictionContactModel/frictionContactModel.C	+0 -59	-59 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/frictionContactModel/frictionContactModel.H	+0 -215	-215 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/frictionContactModel/newFrictionContactModel.C	+0 -83	-83 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/regularizedStandardPenaltyFriction/regularizedStandardPenaltyFriction.C	+0 -165	-165 at 0

src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/regularizedStandardPenaltyFriction/regularizedStandardPenaltyFriction.H	+0 -159	-159 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/slipPenaltyFriction/slipPenaltyFriction.C	+0 -201	-201 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/slipPenaltyFriction/slipPenaltyFriction.H	+0 -204	-204 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/stabilisedPenaltyFriction/stabilisedPenaltyFriction.C	+0 -358	-358 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/stabilisedPenaltyFriction/stabilisedPenaltyFriction.H	+0 -217	-217 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/standardPenaltyFriction/standardPenaltyFriction.C	+0 -135	-135 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/standardPenaltyFriction/standardPenaltyFriction.H	+0 -156	-156 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/tanhPenaltyFriction/tanhPenaltyFriction.C	+0 -154	-154 at 0
src/wireBunchingModels/beamContactModel/contactModels/frictionContactModels/tanhPenaltyFriction/tanhPenaltyFriction.H	+0 -167	-167 at 0
src/wireBunchingModels/beamContactModel/contactModels/normalContactModels/lagrangeMultipliers/lagrangeMultipliers.C	+0 -111	-111 at 0
src/wireBunchingModels/beamContactModel/contactModels/normalContactModels/lagrangeMultipliers/lagrangeMultipliers.H	+0 -132	-132 at 0
... 21 more code files in src/wireBunchingModels/beamContactModel/contactModels (7), src/wireBunchingModels/beamContactModel/lineContact (7), src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam (3), src/wireBunchingModels/beamContactModel/pointContact (2), src/wireBunchingModels/beamContactModel/toroidalPulleyContact (2)	+0 -10791	

Removed plasticity calculations code

Deletes the unused plasticity stress-resultant return-mapping models.

- beamModels/plasticityStressResultantReturnMethods/: axialForce3dBending, linearElasticMisesPlastic and the plasticityStressResultantReturn base removed

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/axialForce3dBending/#axialForce3dBending.H#	+0 -204	-204 at 0
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/axialForce3dBending/axialForce3dBending.C	+0 -1046	-1046 at 0
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/axialForce3dBending/axialForce3dBending.H	+0 -224	-224 at 0
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/linearElasticMisesPlastic/linearElasticMisesPlastic.C	+0 -1210	-1210 at 0
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/linearElasticMisesPlastic/linearElasticMisesPlastic.H	+0 -282	-282 at 0
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/plasticityStressResultantReturn/newPlasticityStressResultantReturn.C	+0 -84	-84 at 0
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/plasticityStressResultantReturn/plasticityStressResultantReturn.C	+0 -66	-66 at 0
src/wireBunchingModels/beamModels/plasticityStressResultantReturnMethods/plasticityStressResultantReturn/plasticityStressResultantReturn.H	+0 -213	-213 at 0

Remove unused code for plasticity and boundary condition update

Deletes two unused include fragments from the solver directory.

- coupledTotalLagNewtonRaphsonBeam: correctPlasticity.H and correctBoundaryForcesAndMoments.H removed

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/correctBoundaryForcesAndMoments.H	+0 -51	-51 at 0
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/correctPlasticity.H	+0 -131	-131 at 0

Removing unused fvPatchFields and modified Make folder accordingly

Deletes unused or untested boundary conditions and removes them from Make/files. It also removes forceBeamDisplacement from Make/files, although that BC's sources remain until d11a935.

- Removed: axialForceTransverseDisplacement(NR), extrapolatedBeamRotation, fixedPathDisplacement, fixedPathRotation, followerDisplacement, momentBeamRotation, myFollowerForceBeamDisplacementNR
- Make/files: entries for the removed BCs and forceBeamDisplacement deleted

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+0 -8	-1 at 19, -6 at 21, -1 at 22
src/wireBunchingModels/beamModels/fvPatchFields/axialForceTransverseDisplacement/axialForceTransverseDisplacementFvPatchVectorField.C	+0 -321	-321 at 0
src/wireBunchingModels/beamModels/fvPatchFields/axialForceTransverseDisplacement/axialForceTransverseDisplacementFvPatchVectorField.H	+0 -198	-198 at 0
src/wireBunchingModels/beamModels/fvPatchFields/axialForceTransverseDisplacementNR/axialForceTransverseDisplacementNRfVvPatchVectorField.C	+0 -354	-354 at 0
src/wireBunchingModels/beamModels/fvPatchFields/axialForceTransverseDisplacementNR/axialForceTransverseDisplacementNRfVvPatchVectorField.H	+0 -207	-207 at 0
src/wireBunchingModels/beamModels/fvPatchFields/extrapolatedBeamRotation/extrapolatedBeamRotationFvPatchVectorField.C	+0 -181	-181 at 0
src/wireBunchingModels/beamModels/fvPatchFields/extrapolatedBeamRotation/extrapolatedBeamRotationFvPatchVectorField.H	+0 -171	-171 at 0
src/wireBunchingModels/beamModels/fvPatchFields/fixedPathDisplacement/fixedPathDisplacementFvPatchVectorField.C	+0 -263	-263 at 0
src/wireBunchingModels/beamModels/fvPatchFields/fixedPathDisplacement/fixedPathDisplacementFvPatchVectorField.H	+0 -185	-185 at 0
src/wireBunchingModels/beamModels/fvPatchFields/fixedPathRotation/fixedPathRotationFvPatchVectorField.C	+0 -263	-263 at 0
src/wireBunchingModels/beamModels/fvPatchFields/fixedPathRotation/fixedPathRotationFvPatchVectorField.H	+0 -185	-185 at 0
src/wireBunchingModels/beamModels/fvPatchFields/followerDisplacement/#fixedDisplacementFvPatchVectorField.H#	+0 -179	-179 at 0
src/wireBunchingModels/beamModels/fvPatchFields/followerDisplacement/followerDisplacementFvPatchVectorField.C	+0 -266	-266 at 0
src/wireBunchingModels/beamModels/fvPatchFields/followerDisplacement/followerDisplacementFvPatchVectorField.H	+0 -183	-183 at 0
src/wireBunchingModels/beamModels/fvPatchFields/momentBeamRotation/momentBeamRotationFvPatchVectorField.C	+0 -271	-271 at 0
src/wireBunchingModels/beamModels/fvPatchFields/momentBeamRotation/momentBeamRotationFvPatchVectorField.H	+0 -174	-174 at 0
src/wireBunchingModels/beamModels/fvPatchFields/myFollowerForceBeamDisplacementNR/myFollowerForceBeamDisplacementNRfVvPatchVectorField.C	+0 -360	-360 at 0
src/wireBunchingModels/beamModels/fvPatchFields/myFollowerForceBeamDisplacementNR/myFollowerForceBeamDisplacementNRfVvPatchVectorField.H	+0 -186	-186 at 0

beamFoam d11a935 2025-08-27 Seevani Bali

Removing unused fvPatchField forceBeamDisplacement

Deletes the source files of the forceBeamDisplacement boundary condition, which 24ef6de had already removed from the build.

- fvPatchFields/forceBeamDisplacement: .C and .H removed

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/fvPatchFields/forceBeamDisplacement/forceBeamDisplacementFvPatchVectorField.C	+0 -277	-277 at 0
src/wireBunchingModels/beamModels/fvPatchFields/forceBeamDisplacement/forceBeamDisplacementFvPatchVectorField.H	+0 -174	-174 at 0

beamFoam 3c9e06f 2025-08-27 Seevani Bali

Change AUTO_WRITE to NO_WRITE for the variables whose output is not required

Stops writing most internal solver fields to time directories by switching them to NO_WRITE; U, Accl and Omega are among them. Also removes the unused Mref field and comment-cludes headers of deleted BCs.

- beamModel.C: beamIndicator, cellIndicator, L set to NO_WRITE
- coupledTotalLagNewtonRaphsonBeam.C: about 35 fields (U, Accl, Omega, dotOmega, DW, DTheta, Lambda/Lambdaf, ref*, CQ*/CM*, explicit* ...) set to NO_WRITE
- Mref_ member removed; #includes for removed BC headers commented out

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+3 -3	157, 174, 274
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+39 -52	35, 37, 40-42, 101, 114, 128, 141, 166, 180, 193 (+28 more)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+0 -3	-3 at 118

beamFoam 40b7e3a 2025-08-27 Seevani Bali

Comment out unused and un-tested BCs from coupledTotalLagNewtonRaphsonBeamBoundaryConditions.C

Comments out the untested axialForceTransverseDisplacementNR branch in the solver's boundary-condition coupling. The remaining supported BCs are forceBeamDisplacementNR, followerForceBeamDisplacementNR, fixedValue and momentBeamRotationNR.

- coupledTotalLagNewtonRaphsonBeamBoundaryConditions.C: axialForceTransverseDisplacementNR block, including its nested rotation handling, commented out; rest reformatted

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamBoundaryConditions.C	+390 -461	36, 38, 41-43, 156-338, -63 at 745, 747-751, 753-754, 756-766, 768-789, 879-1033 (+1 more)

beamFoam 4934cd2 2025-08-27 Seevani Bali

Remove unused pulleys code

Deletes the unused conicalPulley and toroidalPulley geometry classes.

- beamModel/pulleys/: conicalPulley and toroidalPulley .C/.H removed

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/pulleys/conicalPulley/conicalPulley.C	+0 -387	-387 at 0
src/wireBunchingModels/beamModels/beamModel/pulleys/conicalPulley/conicalPulley.H	+0 -450	-450 at 0
src/wireBunchingModels/beamModels/beamModel/pulleys/toroidalPulley/toroidalPulley.C	+0 -367	-367 at 0
src/wireBunchingModels/beamModels/beamModel/pulleys/toroidalPulley/toroidalPulley.H	+0 -369	-369 at 0

beamFoam e60943b 2025-08-27 Seevani Bali

Renamed fvPatchField myFixedRotation to fixedRotation

Renames the myFixedRotation boundary condition to fixedRotation, including its class, files and type name. Cases that use 'myFixedRotation' must be updated.

- fvPatchFields/fixedRotation: TypeName("fixedRotation"); Make/files updated

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+1 -1	22
src/wireBunchingModels/beamModels/fvPatchFields/{myFixedRotation/myFixedRotationFvPatchVectorField.C => fixedRotation/fixedRotationFvPatchVectorField.C}	+21 -21	rename only
src/wireBunchingModels/beamModels/fvPatchFields/{myFixedRotation/myFixedRotationFvPatchVectorField.H => fixedRotation/fixedRotationFvPatchVectorField.H}	+17 -17	rename only

beamFoam e1220d6 2025-08-27 Seevani Bali

Changed the solution residual to normalised solution residuals to make them scale independent

Normalises the Newton-Raphson solution residuals for W and Theta, making convergence scale-independent. The linear solver's true residual is printed separately, and the material and contact residual checks are dropped.

- Evolve.C: WResidual and ThetaResidual divided by a normalising denominator; prints 'Normalised Solution Residual'
- eigenSolver residual now printed as 'True Equation Residual' and no longer overwrites currentResidual
- materialTol, currentMaterialResidual and curContactResidual commented out of the loop condition

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+39 -35	70-77, 86, 90, 93, 580-588, 602-607, 623, -1 at 713, 853, 856 (+8 more)

beamFoam ce6e4a1 2025-08-27 Seevani Bali

Remove beamsBunchingModel class because it has contact portions and they are not used

Deletes the beamsBunchingModel class, which depended on the removed contact code.

- src/wireBunchingModels/beamsBunchingModel: .C and .H removed

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamsBunchingModel/beamsBunchingModel.C	+0 -789	-789 at 0
src/wireBunchingModels/beamsBunchingModel/beamsBunchingModel.H	+0 -198	-198 at 0

beamFoam deebde1 2025-08-27 Seevani Bali

Tutorial Case: In-plane bending of cantilever into circle - IJNME - Test case 1

Adds tutorials/cantilever: in-plane bending of a cantilever into a circle by an end moment (IJNME test case).

File	Lines	Changed line ranges (after the commit)
tutorials/cantilever/Allclean	+10 -0	1-10
tutorials/cantilever/Allrun	+10 -0	1-10
tutorials/cantilever/ (case files)	+463 -0	12 files

beamFoam 1c8429a 2025-08-27 Seevani Bali

Test case: In-plane quasi-static deformation of cantilever subjected to follower end force - Test case 2 of OFW paper

Adds tutorials/cantileverFollowerForce: quasi-static in-plane cantilever under a follower end force (OFW paper test case 2).

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverFollowerForce/ (case files)	+357 -0	9 files

beamFoam 98f81ed 2025-08-27 Seevani Bali

Test Case: 3D dynamic cantilever column subjected to tip force - Test Case 1 of OFW paper

Adds tutorials/3DdynamicCantilever: dynamic 3D cantilever column under a tip force (OFW paper test case 1).

File	Lines	Changed line ranges (after the commit)
tutorials/3DdynamicCantilever/Allclean	+10 -0	1-10
tutorials/3DdynamicCantilever/Allrun	+13 -0	1-13
tutorials/3DdynamicCantilever/ (case files)	+401 -0	10 files

beamFoam 5fcc421 2025-08-27 Seevani Bali

Adding Allclean and Allrun files to cantileverFollowerForce tutorial case

Adds Allrun and Allclean scripts to the cantileverFollowerForce tutorial.

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverFollowerForce/Allclean	+10 -0	1-10
tutorials/cantileverFollowerForce/Allrun	+10 -0	1-10

beamFoam 349bb93 2025-08-27 Seevani Bali

Test Case : 3D flexible beam with free edges subjected to force and moment at end - Test Case 3 of OFW paper

Adds tutorials/freeFlexibleBeam: 3D free-free flexible beam driven by end forces and moments (OFW paper test case 3).

File	Lines	Changed line ranges (after the commit)
tutorials/freeFlexibleBeam/Allclean	+10 -0	1-10
tutorials/freeFlexibleBeam/Allrun	+12 -0	1-12
tutorials/freeFlexibleBeam/ (case files)	+446 -0	12 files

beamFoam 27f789a 2025-08-29 Seevani Bali

Changes the convergence loop to include residual, solution and absolute tolerance checks

Replaces the Newton-Raphson convergence test with a PETSc SNES-style check: absolute residual, relative residual, relative step and divergence tolerances plus a maximum iteration count. Also moves the post-solve field update into its own function.

- coupledTotalLagNewtonRaphsonBeam.H: new checkConvergence(...) and updateSolutionVariables()
- Evolve.C: new beamProperties keywords residualTol (1e-8), solutionTol (1e-10), absoluteTol (1e-30), divergenceTol (1e4), infoFrequency
- Prints a residual table and the reason for convergence or divergence; evolve() returns the initial residual norm

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+51 -3	-2 at 262, 293-343
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+337 -348	40, 64-77, 79, -1 at 80, 82-91, 98-101, -1 at 103, -30 at 110, -3 at 113, 156 (+17 more)

beamFoam 5cae30a 2025-08-29 Seevani Bali

Amended beamProperties file in cantilever follower load test case

Updates the cantileverFollowerForce beamProperties: removes the second beam entry, uses 40 segments, adopts the new solutionTol/residualTol keywords and turns debug off.

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverFollowerForce/ (case files)	+4 -21	1 file

Removed madatory lookups of start and end patch name from tets cases - modified beamModel.C base class file

Stops reading startPatchName and endPatchName from beamProperties and hard-codes them to 'left' and 'right'. Only beamModel.C changes; no test-case files are touched, despite what the message says.

- beamModel.C: lookup of startPatchName/endPatchName replaced by fixed "left"/"right"

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+5 -3	583-587

Removed deprecated mesh utilities createCircularBeamMesh and createMultipleBeamMesh

Deletes the deprecated createCircularBeamMesh and createMultipleBeamMesh utilities, including committed Make build artefacts, leaving createBeamMesh as the only mesh generator.

- applications/utilities/createCircularBeamMesh and createMultipleBeamMesh removed

File	Lines	Changed line ranges (after the commit)
applications/utilities/createCircularBeamMesh/Make/files	+0 -4	-4 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/dependencies	+0 -1	-1 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/dependencyFiles	+0 -2	-2 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/dontIncludeDeps	+0 -0	
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/files	+0 -60	-60 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/filesMacros	+0 -1	-1 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/includeDeps	+0 -1	-1 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/localObjectFiles	+0 -2	-2 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/objectFiles	+0 -2	-2 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/options	+0 -62	-62 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDPDebug/sourceFiles	+0 -2	-2 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/dependencies	+0 -1	-1 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/dependencyFiles	+0 -2	-2 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/dontIncludeDeps	+0 -0	
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/files	+0 -60	-60 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/filesMacros	+0 -1	-1 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/includeDeps	+0 -1	-1 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/localObjectFiles	+0 -2	-2 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/objectFiles	+0 -2	-2 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/options	+0 -62	-62 at 0
applications/utilities/createCircularBeamMesh/Make/linux64GccDP0pt/sourceFiles	+0 -2	-2 at 0
applications/utilities/createCircularBeamMesh/Make/options	+0 -9	-9 at 0
applications/utilities/createCircularBeamMesh/createCircularBeamMesh.C	+0 -335	-335 at 0
applications/utilities/createCircularBeamMesh/printMeshSummary.H	+0 -85	-85 at 0
applications/utilities/createMultipleBeamMesh/Make/files	+0 -4	-4 at 0
applications/utilities/createMultipleBeamMesh/Make/options	+0 -13	-13 at 0
applications/utilities/createMultipleBeamMesh/createMultipleBeamMesh.C	+0 -587	-587 at 0

beamFoam 992731d 2025-08-29 Seevani Bali

Add printMeshSummary.H to createBeamMesh utility

Adds printMeshSummary.H to createBeamMesh; the file is identical to the one deleted with createCircularBeamMesh.

- applications/utilities/createBeamMesh/printMeshSummary.H added

File	Lines	Changed line ranges (after the commit)
applications/utilities/createBeamMesh/printMeshSummary.H	+85 -0	1–85

beamFoam d237ba0 2025-08-29 Philip Cardiff

Tidy up and output from coupledTotalLagNewtonRaphsonBeam and remove duplicate debug flag

Removes a local 'debug' variable read from beamProperties that duplicated the existing debug flag, and prints the total iteration count only when debug is on.

- coupledTotalLagNewtonRaphsonBeamEvolve.C: duplicate debug lookup removed; 'Total iterations' output gated by debug

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+5 -7	-5 at 92, -1 at 637, 639–643

beamFoam fc21172 2025-08-31 Amirhossein Taran

adding github workflows to the repo

Adds a GitHub Actions workflow that builds beamFoam with Allwmake in an opencfd/openfoam-dev:2312 container on push, pull request or manual dispatch.

- .github/workflows/build.yml added

File	Lines	Changed line ranges (after the commit)
.github/workflows/build.yml	+42 -0	1–42

beamFoam 240d493 2025-08-31 Amirhossein Taran

Optimize build command in GitHub Actions workflow to use parallel compilation

Builds with './Allwmake -j' in the CI workflow so compilation runs in parallel.

- .github/workflows/build.yml: Allwmake -> Allwmake -j

File	Lines	Changed line ranges (after the commit)
.github/workflows/build.yml	+1 -1	42

beamFoam 740f36f 2025-08-31 Amirhossein Taran

Clarify step name in GitHub Actions workflow for compiling beamFoam

Renames the CI build step to 'Source OpenFOAM environment and compiling beamFoam'.

File	Lines	Changed line ranges (after the commit)
.github/workflows/build.yml	+1 -1	35

beamFoam 05a1e59 2025-09-01 Seevani Bali

Adding README files to 3DdynamicCantilever + changes to Allrun for plotting

Adds a README and an allPlots.gnuplot script to the 3DdynamicCantilever tutorial, hooks plotting into Allrun, and updates beamProperties (40 segments, solutionTol, patch-name and Newmark keywords removed).

File	Lines	Changed line ranges (after the commit)
tutorials/3DdynamicCantilever/Allrun	+12 -0	8, 11, 15, 17–25
tutorials/3DdynamicCantilever/allPlots.gnuplot	+48 -0	1–48
tutorials/3DdynamicCantilever/README.md	+78 -0	document
tutorials/3DdynamicCantilever/ (case files)	+3 -10	1 file

beamFoam ae18c78 2025-09-01 Seevani Bali

Adding gnuplots for result plotting in 3D dynamic cantilever case

Adds gnuplot scripts to the 3DdynamicCantilever tutorial for mesh and time-step convergence plots, and adjusts Allrun and Allclean.

File	Lines	Changed line ranges (after the commit)
tutorials/3DdynamicCantilever/Allclean	+2 -0	11-12
tutorials/3DdynamicCantilever/Allrun	+2 -3	1, 12
tutorials/3DdynamicCantilever/allPlots.gnuplot	+7 -7	9-10, 24-25, 27, 32-33
tutorials/3DdynamicCantilever/dispPlotDT.gnuplot	+31 -0	1-31
tutorials/3DdynamicCantilever/dispPlotMesh.gnuplot	+30 -0	1-30

beamFoam a9c4091 2025-09-01 Seevani Bali

Reference results + shell scripts for running varying mesh + varying dt

Adds reference results (an Abaqus curve and a solid-model point-displacement file) plus runSweep.sh and runTimeSweep.sh, which rerun the 3DdynamicCantilever case over several mesh sizes and time steps.

File	Lines	Changed line ranges (after the commit)
tutorials/3DdynamicCantilever/runSweep.sh	+65 -0	1-65
tutorials/3DdynamicCantilever/runTimeSweep.sh	+67 -0	1-67
tutorials/3DdynamicCantilever/cantilever_abaqus_50percent.txt	+1002 -0	document
tutorials/3DdynamicCantilever/ (case files)	+271 -0	1 file

beamFoam acab145 2025-09-02 Seevani Bali

3D dynamic cantilever tutorial case - minor change sto plotting and add comments to fvSchemes

Tweaks the 3DdynamicCantilever plot and time-sweep scripts. It also changes the case's time schemes from Newmark to Euler, not only adding comments, and documents optional newmarkBeta(W)/newmarkGamma(W) values.

File	Lines	Changed line ranges (after the commit)
tutorials/3DdynamicCantilever/dispPlotDT.gnuplot	+2 -1	24-25
tutorials/3DdynamicCantilever/runTimeSweep.sh	+6 -2	4-9
tutorials/3DdynamicCantilever/ (case files)	+10 -2	1 file

beamFoam 0712571 2025-09-02 Seevani Bali

Add gnuplot + readme to test case 3 - free flexible beam

Adds a README, an allPlots.gnuplot script and Allrun plotting steps to the freeFlexibleBeam tutorial.

File	Lines	Changed line ranges (after the commit)
tutorials/freeFlexibleBeam/Allclean	+2 -0	11-12
tutorials/freeFlexibleBeam/Allrun	+9 -0	8, 11, 14, 16-21
tutorials/freeFlexibleBeam/allPlots.gnuplot	+51 -0	1-51
tutorials/freeFlexibleBeam/README.md	+71 -0	document

beamFoam 2698b42 2025-09-02 Seevani Bali

minor change solution tolerance to free flex beam case

Relaxes freeFlexibleBeam solutionTol from 1e-13 to 1e-10 and removes the now-unused startPatchName/endPatchName entries.

File	Lines	Changed line ranges (after the commit)
tutorials/freeFlexibleBeam/ (case files)	+2 -5	1 file

beamFoam 3ea51a1 2025-09-02 Seevani Bali

Add Readme + reference files + plot files to cantilever follower force

Adds a README, Simo reference results, a displacement plot script and a mesh-sweep script to cantileverFollowerForce, and updates its beamProperties tolerances and patch-name keywords.

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverFollowerForce/Allclean	+3 -1	10–12
tutorials/cantileverFollowerForce/dispPlot.gnuplot	+40 -0	1–40
tutorials/cantileverFollowerForce/runSweep.sh	+69 -0	1–69
tutorials/cantileverFollowerForce/README.md	+60 -0	document
tutorials/cantileverFollowerForce/ (case files)	+57 -4	2 files

beamFoam 585f6c3 2025-09-02 Seevani Bali

Bug Fix in Euler inertia term additions to the source terms

Fixes the explicit inertia source for the Euler scheme: it now uses fvc::ddt(U) and fvc::ddt(Omega) instead of the Newmark-only Accl and dotOmega fields. Newmark keeps its previous terms.

- coupledTotalLagNewtonRaphsonBeamEvolve.C: QRho/MRho built per d2dt2 scheme
- updateSolutionVariables(): Euler no longer assigns Accl_ = ddt(U) or dotOmega_ = ddt(Omega)

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+49 -11	342–343, 345–346, 348–357, 360–392, 808, 882

beamFoam 2bc84b3 2025-09-03 Seevani Bali

Added plotting to cantilever follower force + README

Adds load-history data and an allPlots.gnuplot script to cantileverFollowerForce, plots from Allrun and revises its README.

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverFollowerForce/Allrun	+8 -0	8, 11, 13–18
tutorials/cantileverFollowerForce/allPlots.gnuplot	+28 -0	1–28
tutorials/cantileverFollowerForce/README.md	+13 -17	document
tutorials/cantileverFollowerForce/ (case files)	+1001 -0	1 file

beamFoam bd3d0c7 2025-09-03 Seevani Bali

modified README

Revises the 3DdynamicCantilever README. It also switches the case's ddt/d2dt2 schemes back from Euler to Newmark, which the message ('modified README') does not mention.

File	Lines	Changed line ranges (after the commit)
tutorials/3DdynamicCantilever/README.md	+20 -17	document
tutorials/3DdynamicCantilever/ (case files)	+2 -2	1 file

beamFoam d1bd069 2025-09-03 Seevani Bali

modified REAdMe in free flexible beam case

Shortens and edits the freeFlexibleBeam README.

File	Lines	Changed line ranges (after the commit)
tutorials/freeFlexibleBeam/README.md	+7 -16	document

beamFoam 5a07ce4 2025-09-03 Seevani Bali

Added README to cantilever case

Adds a README to the cantilever tutorial describing the end-moment bending benchmark, its setup and its expected results.

File	Lines	Changed line ranges (after the commit)
tutorials/cantilever/README.md	+63 -0	document

beamFoam f216707 2025-09-03 Seevani Bali

Modified global README.md file

Updates the top-level README: notes that each case has a README, adds gnuplot plotting instructions and removes outdated paths and the contact section.

File	Lines	Changed line ranges (after the commit)
README.md	+5 -21	document

2.2 November – December 2025: momentum contributions, spring boundary condition, build fixes

Also part of the `openfoam-v2306-seevani` work. The first `moorFV` commit (local changes) falls in this period.

beamFoam 1b31789 2025-11-14 Seevani Bali

Updates to cantilever test case

Updates the cantilever tutorial. Allrun now plots tip displacement with gnuplot and opens the PDF, a plot script and output PDF are added, and the time step drops from 0.2 to 0.01.

File	Lines	Changed line ranges (after the commit)
tutorials/cantilever/Allrun	+4 -0	11-14
tutorials/cantilever/allPlots.gnuplot	+25 -0	1-25
tutorials/cantilever/displacementPlot.pdf	bin	document
tutorials/cantilever/ (case files)	+1 -1	1 file

Add momentumContribution structure with empty morisonMomentum

Adds a run-time-selectable momentumContribution base class and hooks it into the coupled Newton-Raphson beam solver, so external linear and angular momentum sources plus diagonal coefficients can be added per cell. The morisonMomentum stub only returns zero fields.

- New abstract momentumContribution (an IOdictionary) with diagCoeff, linearMomentumSource and angularMomentumSource, read from constant/momentumContributionProperties
- coupledTotalLagNewtonRaphsonBeam gains autoPtr momentumContribPtr_; evolve() adds the sources to source rows 0-5 and the coefficients to d
- The header check is inverted (!typeHeaderOk), so the object is created only when the file is missing
- morisonMomentum.C is not added to Make/files, so it is not compiled

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+2 -0	29-30
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+21 -0	640, 1071-1090
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+7 -4	55, 87, 134, 227-230
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+37 -0	589-625
src/wireBunchingModels/momentumContribution/momentumContribution.C	+116 -0	1-116
src/wireBunchingModels/momentumContribution/momentumContribution.H	+162 -0	1-162
src/wireBunchingModels/momentumContribution/morisonMomentum/morisonMomentum.C	+120 -0	1-120
src/wireBunchingModels/momentumContribution/morisonMomentum/morisonMomentum.H	+132 -0	1-132

Updates to momentumContribution base class and derived morisonMomentum class

Moves the Morison drag calculation out of the beam solver's evolve() into morisonMomentum, and changes the contribution interface to take the beam model and velocity field instead of W and Theta.

- Removes the inline dragActive_ drag block and its steady-state check from coupledTotalLagNewtonRaphsonBeamEvolve.C
- morisonMomentum reads Cdn/Cdt from <type>Coeffs and computes normal and tangential drag along the HermiteSpline tangents
- Raises a FatalError if ddt(W) uses steadyState
- Signatures change to (const beamModel&, const volVectorField& U); morisonMomentum.C is added to Make/files

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+1 -0	30
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+1 -1	495
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+83 -81	504-508, 511-564, 569-587, 596-597, 602-603, 621
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamMatrix.C	+1 -1	1
src/wireBunchingModels/momentumContribution/momentumContribution.H	+16 -10	53, -1 at 113, 129-130, 133-135, 139-140, 143-145, 149-150, 153-155
src/wireBunchingModels/momentumContribution/morisonMomentum/morisonMomentum.C	+117 -19	48-59, 67-69, 72, 79-82, 92-94, 98-108, 112-190, 198-200, 203
src/wireBunchingModels/momentumContribution/morisonMomentum/morisonMomentum.H	+29 -19	26, 29-30, 37, 46-48, 55-56, 69-70, 72-73, 99-100, 103-105, 109-110 (+4 more)

Base beamContribution class with some working examples - groundContact - not working properly

Replaces momentumContribution with beamMomentumContribution, which allows a list of contributions per case. Morison drag becomes morisonDragContribution and the inline ground-contact code becomes groundContactContribution. The commit message notes ground contact did not yet work properly.

- beamMomentumContribution is built by name from a dictionary; the solver reads a beamMomentumContributions list from constant/beamMomentumContributionProperties into a PtrList
- evolve() loops over every contribution, adding its sources and diagonal coefficients
- groundContact applies a z-direction penalty spring/damper below groundZ, using optional refW from 0/, and prints debug output
- Removes the dragActive_/Cdn_/Cdt_ members and the inline groundContactActive_ block from the solver

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+3 -2	29–31
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+1 -1	142
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+37 -11	640, 642–645, 1073, 1075, 1080–1081, 1083, 1085–1110, 1114
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+10 -10	55, 227–228, 230–232, 234–235, 237–238
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+47 -46	287–292, 294–303, 589–590, 592–598, 600–604, 606–610, 612–615, 617, 619–622, 624–625 (+1 more)
src/wireBunchingModels/beamMomentumContribution/beamMomentumContribution.C	+149 -0	1–149
src/wireBunchingModels/{momentumContribution/momentumContribution.H => beamMomentumContribution/beamMomentumContribution.H}	+26 -25	rename only
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.C	+208 -0	1–208
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.H	+148 -0	1–148
src/wireBunchingModels/{momentumContribution/morisonMomentum/morisonMomentum.C => beamMomentumContribution/morisonDragContribution/morisonDragContribution.C}	+21 -12	rename only
src/wireBunchingModels/{momentumContribution/morisonMomentum/morisonMomentum.H => beamMomentumContribution/morisonDragContribution/morisonDragContribution.H}	+21 -19	rename only
src/wireBunchingModels/momentumContribution/momentumContribution.C	+0 -116	–116 at 0

Working tutorial - cantileverDa=rag to test morisonDrag contribution

Adds the cantileverDrag tutorial, which tests the morisonDrag contribution, with case files, a README, a gnuplot script and plots with and without drag.

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverDrag/Allclean	+10 -0	1–10
tutorials/cantileverDrag/Allrun	+14 -0	1–14
tutorials/cantileverDrag/allPlots.gnuplot	+25 -0	1–25
tutorials/cantileverDrag/README.md	+63 -0	document
tutorials/cantileverDrag/displacementPlot.pdf	bin	document
tutorials/cantileverDrag/displacementPlotWithMorisonDrag.pdf	bin	document
tutorials/cantileverDrag/displacementPlotWithNoDrag.pdf	bin	document
tutorials/cantileverDrag/ (case files)	+528 -0	13 files

Added Acceleration as an input to the memmber functions of the beamMomentumContribution class - needed for added mass terms

Adds the acceleration field `Accl` as a third argument to every `beamMomentumContribution` member function, so later added-mass terms can use it. The solver now passes `Accl_` to each contribution.

- `diagCoeff`, `linearMomentumSource` and `angularMomentumSource` now take `(bm, U, Accl)` in the base class, `morisonDragContribution` and `groundContactContribution`
- Removes commented-out leftover code from `beamMomentumContribution.C`

File	Lines	Changed line ranges (after the commit)
<code>src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C</code>	+3 -3	597, 603, 621
<code>src/wireBunchingModels/beamMomentumContribution/beamMomentumContribution.C</code>	+0 -43	-23 at 48, -9 at 70, -11 at 90
<code>src/wireBunchingModels/beamMomentumContribution/beamMomentumContribution.H</code>	+6 -6	135-136, 145-146, 155-156
<code>src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.C</code>	+8 -8	82-83, 107-108, 156, 158, 189-190
<code>src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.H</code>	+9 -5	109-110, 114-115, 124-126, 134-135
<code>src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.C</code>	+6 -6	77-78, 102-103, 208-209
<code>src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.H</code>	+9 -4	105, 109-110, 116, 120-121, 127, 131-132

Add a fatal error if inconsistent `d2dt2Scheme` is used and uncomment `Accl` field for Euler method because it may be required for other force contributions

Adds a `FatalError` when the `d2dt2Scheme` is not recognised during source assembly. It also enables the `Accl_ = fvc::ddt(U_)` update for the Euler scheme so contributions get a valid acceleration.

- Adds an else branch with `FatalErrorInFunction` for an undefined `d2dt2SchemeName_` in `coupledTotalLagNewtonRaphsonBeamEvolve.C`
- Uncomments `Accl_ = fvc::ddt(U_)` in `updateSolutionVariables()` for Euler

File	Lines	Changed line ranges (after the commit)
<code>src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C</code>	+7 -1	392-397, 814

Alignment fixes + remove legacy code and unwanted comments + add author

Cleanup with no change in behaviour: removes the leftover ground-contact members from the beam solver, deletes commented legacy code, fixes indentation and adds an author credit.

- Removes groundContactActive_, gDamping_, gStiffness_ and groundZ_ from coupledTotalLagNewtonRaphsonBeam
- Removes debug Info prints from groundContactContribution and notes that contact is z-direction only
- Adds Amirhossein Taran as author in the contribution headers
- Removes commented blocks from the cantileverDrag beamMomentumContributionProperties

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+1 -12	640, -12 at 641
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+0 -24	-24 at 229
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+2 -108	-19 at 285, -62 at 483, -22 at 485, 493, 498, -1 at 512
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamMatrix.C	+1 -1	1
src/wireBunchingModels/beamMomentumContribution/beamMomentumContribution.C	+4 -4	74-77
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.C	+11 -10	64-65, 68, -1 at 69, 126, 135, 155-156, 158, 165-166, 174
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.H	+2 -0	32, 40
src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.C	+3 -15	87, 92-93, -12 at 168
src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.H	+1 -0	38
tutorials/cantileverDrag/ (case files)	+0 -29	1 file

Add rhoFluid parameter to morisonDragContribution for fluid density handling

morisonDragContribution now reads a rhoFluid coefficient and uses the fluid density in the drag force instead of the beam density.

- Fdn and Fdt use rhoFluid_ in place of bm.rho()
- The rhoFluid_ member is declared in the header only in the next commit (95ad89d), so this commit alone would not compile

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.C	+6 -2	65-68, 170-171

beamFoam 95ad89d 2025-11-19 Amir Taran

Refactor ground contact model: update coefficients and add tangential friction handling

Rewrites the ground-contact force with separate normal stiffness and damping, reverses the sign convention for the normal force, and adds stick-slip Coulomb tangential friction. This revision does not compile.

- Replaces gDamping/gStiffness with kNormal_, cNormal_, kTangential_ and muFriction_
- Tangential force is a linear spring capped at muFriction*|normal force|
- Wrong lookup keys ("kNormal_" is read three times), mismatched member names and an undeclared spline
- Also declares rhoFluid_ in morisonDragContribution.H

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.C	+43 -11	58, 60, 62, 64–72, 99, 104, -2 at 172, 174–186, 194–209
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.H	+10 -4	75–76, 78–85
src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.H	+3 -0	79–81

beamFoam f1bd715 2025-11-20 Seevani Bali

Corrected compilation errors in groundContactContribution + removed unused beam.rho form morisonDrag

Fixes the compile errors in groundContactContribution from the previous commit and removes the unused beam density variable from morisonDragContribution.

- Correct lookup keys: kNormal, cNormal, kTangential, muFriction
- Member names now match the header (cNormal_, kTangential_)
- Builds the HermiteSpline in linearMomentumSource and adds UtHat for the friction direction
- Removes const scalar rho = bm.rho().value() from morisonDragContribution.C

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.C	+31 -10	60, 64, 66, 68, 72, 174–180, 193–195, 197, 201, 208 (+4 more)
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.H	+2 -2	79, 82
src/wireBunchingModels/beamMomentumContribution/morisonDragContribution/morisonDragContribution.C	+1 -1	165

beamFoam c599cf0 2025-11-20 Seevani Bali

Style changes for consistency

Style-only changes in groundContactContribution: the normal force becomes const and the friction condition loses its redundant parentheses.

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.C	+4 -6	205, 209, 213–214

beamFoam 71a9e43 2025-11-27 Seevani Bali

Changes to top-level README.md to ad thirdparty package version and minor changes

README update that documents the Eigen third-party dependency (tested with 3.3.7), names the createBeamMesh utility and fixes typos.

File	Lines	Changed line ranges (after the commit)
README.md	+5 -3	document

beamFoam 8478030 2025-12-08 Seevani Bali

Minor change to dimension of refW in groundContactContribution.C file + add author in followerForce BC

Fixes the dimensions of the default refW field in groundContactContribution from velocity to length, and adds an author line to the followerForceBeamDisplacementNR boundary condition header.

- dimensionedVector("refW", dimLength, ...) replaces dimVelocity
- Adds Seevani Bali as author in followerForceBeamDisplacementNRFvPatchVectorField.H

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/fvPatchFields/followerForceBeamDisplacementNR/followerForceBeamDisplacementNRFvPatchVectorField.H	+3 -0	34–36
src/wireBunchingModels/beamMomentumContribution/groundContactContribution/groundContactContribution.C	+1 -1	157

beamFoam d6d8035 2025-12-08 Seevani Bali

New linear spring boundary condition derived from existing forceBeamDisplacementNR BC is added

Adds the linearSpringForceBeamDisplacementNR boundary condition. It derives from forceBeamDisplacementNR and applies a linear spring force along a chosen direction at a beam end, updating the Newton-Raphson patch coefficients.

- Reads springStiffness and springDirection (normalised) from a linearSpringCoeffs subdictionary; $Ks_ = k * d$
- evaluate() sets force() to $-k (W \cdot d) d$ on each face and changes the CQW/delta coefficient by $Ks_$
- Includes a commented-out placeholder for an extra time-varying force
- Registers the new source file in Make/files

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+1 -0	23
src/wireBunchingModels/beamModels/fvPatchFields/linearSpringForceBeamDisplacementNR/linearSpringForceBeamDisplacementNRFvPatchVectorField.C	+318 -0	1–318
src/wireBunchingModels/beamModels/fvPatchFields/linearSpringForceBeamDisplacementNR/linearSpringForceBeamDisplacementNRFvPatchVectorField.H	+188 -0	1–188

beamFoam 8d1096a 2025-12-08 Seevani Bali

Beam and spring in series in x-direction

Adds the beamAndSpringInSeries tutorial: a beam with a fixed left end and a linearSpringForceBeamDisplacementNR spring at the right end, acting in x.

File	Lines	Changed line ranges (after the commit)
tutorials/beamAndSpringInSeries/Allclean	+10 -0	1–10
tutorials/beamAndSpringInSeries/Allrun	+14 -0	1–14
tutorials/beamAndSpringInSeries/ (case files)	+535 -0	13 files

beamFoam 94fa2f5 2025-12-08 Seevani Bali

Cantilever subjected to udl with a vertical spring at one end

Adds the cantileverSpring tutorial: a cantilever under a uniformly distributed load with a vertical linear spring (stiffness 800, direction y) at the free end. It includes a README, a plot script and a result PDF.

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverSpring/Allclean	+10 -0	1–10
tutorials/cantileverSpring/Allrun	+14 -0	1–14
tutorials/cantileverSpring/allPlots.gnuplot	+25 -0	1–25
tutorials/cantileverSpring/README.md	+63 -0	document
tutorials/cantileverSpring/displacementPlot.pdf	bin	document
tutorials/cantileverSpring/ (case files)	+486 -0	11 files

beamFoam 50fb9cd 2025-12-09 Philip Cardiff

Minor edits to beamAndSpringInSeries

Tidies the beamAndSpringInSeries tutorial: removes commented boundary blocks and the gnuplot/open steps from Allrun, and sets startFrom to startTime.

File	Lines	Changed line ranges (after the commit)
tutorials/beamAndSpringInSeries/Allrun	+0 -4	-4 at 10
tutorials/beamAndSpringInSeries/ (case files)	+3 -22	2 files

beamFoam 3deff39 2025-12-09 Philip Cardiff

Remove old files

Deletes the README, gnuplot scripts, Allrun plot steps and result PDFs from the cantileverDrag and cantileverSpring tutorials. It also adds rhoFluid 1000 to the drag case and shortens its endTime to 0.15.

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverDrag/Allrun	+0 -4	-4 at 10
tutorials/cantileverDrag/allPlots.gnuplot	+0 -25	-25 at 0
tutorials/cantileverSpring/Allrun	+0 -4	-4 at 10
tutorials/cantileverSpring/allPlots.gnuplot	+0 -25	-25 at 0
tutorials/cantileverDrag/README.md	+0 -63	document
tutorials/cantileverDrag/displacementPlot.pdf	bin	document
tutorials/cantileverDrag/displacementPlotWithMorisonDrag.pdf	bin	document
tutorials/cantileverDrag/displacementPlotWithNoDrag.pdf	bin	document
tutorials/cantileverSpring/README.md	+0 -63	document
tutorials/cantileverSpring/displacementPlot.pdf	bin	document
tutorials/cantileverDrag/ (case files)	+7 -6	2 files

beamFoam f175274 2025-12-10 Seevani Bali

Added time-varying offset force feature to spring BC of beamsolver + modified tutorial case

Adds an optional time-varying offset force to the linear spring BC, read from an offsetForceSeries table. It also changes the spring term in the patch coefficient from $-K_s$ to $+K_s$, which is the correct sign for a restoring spring (the code comment explains it). The message does not mention this.

- New offsetForce_ (vectorField) and offsetForceSeries_ (interpolationTable<vector>) members, mapped in autoMap/rmap
- updateCoeffs() re-evaluates offsetForce_ at the current time
- Coefficient changes from $CQW/\delta - K_s$ to $CQW/\delta + K_s$ (a sign fix; not in the commit message)
- Tutorial beamAndSpringInSeries adds an offsetForceSeries entry and constant/timeVsOffsetForce (100 N in x)

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/fvPatchFields/linearSpringForceBeamDisplacementNR/linearSpringForceBeamDisplacementNRFvPatchVectorField.C	+41 -29	50-52, 71-73, 88-90, 126-131, 133-138, 151-153, 167-169, 182, 199, 211-215 (+3 more)
src/wireBunchingModels/beamModels/fvPatchFields/linearSpringForceBeamDisplacementNR/linearSpringForceBeamDisplacementNRFvPatchVectorField.H	+4 -4	71-72, 74-75
tutorials/beamAndSpringInSeries/ (case files)	+10 -0	2 files

beamFoam 725dc50 2025-12-10 Seevani Bali

Removed cantilever drag test case for testing morisonDrag beam contribution

Removes the cantileverDrag tutorial completely.

File	Lines	Changed line ranges (after the commit)
tutorials/cantileverDrag/Allclean	+0 -10	-10 at 0
tutorials/cantileverDrag/Allrun	+0 -10	-10 at 0
tutorials/cantileverDrag/ (case files)	+0 -500	13 files

beamFoam 1882ba1 2025-12-10 Philip Cardiff

Remove unused timeVsForce

Removes the unused constant/timeVsForce file from the beamAndSpringInSeries tutorial.

File	Lines	Changed line ranges (after the commit)
tutorials/beamAndSpringInSeries/ (case files)	+0 -4	1 file

beamFoam 8544eea 2025-12-10 Philip Cardiff

Allwmake: check OpenFOAM is sourced

Allwmake now exits if WM_PROJECT is unset (OpenFOAM not sourced). It warns, without stopping, when the sourced version is not in a list from v2106 to v2506.

File	Lines	Changed line ranges (after the commit)
Allwmake	+13 -0	6-18

beamFoam b4f737a 2025-12-10 Philip Cardiff

Allwmake: use bash instead of sh

Changes the Allwmake shebang from sh to bash, which its [[]] checks need. It also makes the script run from its own directory.

- Adds cd "\${0%/*}" || exit; the commit message does not mention this

File	Lines	Changed line ranges (after the commit)
Allwmake	+2 -1	1-2

moorFV 1ae0db4 2025-12-12 Colm Mc Alister

Save my local changes

Saves local work: the finiteVolumeBeam restraint now writes the beam's anchor-end displacement to a history file each step. Bumps the beamFoam submodule to 3deff39 ('Remove old files'), which predates the 6DOF commits.

- finiteVolumeBeam.C/.H: displacementFilePtr_ writes displacement<name>.dat (W on the anchor patch)
- sDoFRGBFvBeam/Make/options: hard-coded /Users/... include path for beamModel
- src/beamFoam -> 3deff39

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1
src/sDoFRGBFvBeam/Make/options	+2 -1	11-12
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.C	+30 -1	93-94, 161-168, 190-198, 283, 311-320
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.H	+1 -0	90

2.3 February – August 2026: first rigid-body coupling attempt (Colm Mc Alister, with Codex)

Adds rigid-body unknowns to the BlockEigen system and moorFV state passing. The review on 30 September 2026 (rigidBodyCouplingPRSummary.pdf) found that this path is still a partitioned scheme. The descriptions below say what the code does.

beamFoam d45a8d4 2026-02-09 Colm Mc Alister

git check

Adds a single '//git check' comment to BlockEigenSolverOF.C to test the git setup. No functional change.

- BlockEigenSolverOF.C: comment line only

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+3 -0	43-45

beamFoam ebe8931 2026-02-09 Colm Mc Alister

Adding 6x6 block with 1s along diagonal

Grows the Eigen system by six rows and puts a 6x6 identity block in them, with zero right-hand side and zero initial guess. The extra rows are not linked to the beam rows, so the beam solution does not change.

- BlockEigenSolverOF::convertFoamMatrixToEigenMatrix: nRows = 6*(d.size() + 1), 36 extra triplets reserved
- Identity triplets at rbRow = 6*d.size()
- BlockEigenSolverOF::solve: six zero RHS entries; result copied into an unused local vector rigidBodyX
- Removes the d45a8d4 test comment

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+44 -6	-3 at 42, 77-78, 80, 84, 93, 120-138, 260-264, 266, 280-285, 299 (+1 more)

beamFoam 955aa6a 2026-02-12 Colm Mc Alister

adding 6DOFx input/output vector

Passes the six extra solution entries back to the caller through a new solve() argument. The beam model creates a zero 6x1 matrix for it; the result is not used.

- BlockEigenSolverOF::solve gains scalarRectangularMatrix& sixDOFx (.C and .H)
- coupledTotalLagNewtonRaphsonBeamEvolve.C: creates sixDOFx(6,1,0.0) and passes it to eigenSolver.solve
- Residual loop uses nCells = d_.size() + 1

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+4 -1	540-542, 545
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+30 -26	77-78, 81, 93-94, 124, 127, 131-140, 228-230, 264, 267, 270 (+7 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+2 -1	134-135

beamFoam 39386fb 2026-02-16 Colm Mc Alister

Basic 6DOF equations added

Fills the six extra RHS entries with a Newmark-beta displacement and rotation prediction computed from hard-coded placeholder values ($\Delta t = 1$, constant velocity, acceleration and torque). The matrix block stays an identity, so the extra unknowns just equal this RHS.

- BlockEigenSolverOF::solve: rigidStart = nRows - 6; beta = 0.25, gamma = 0.5
- Translational RHS: $d_0 + dt*v_0 + dt^2*(beta*a + (0.5 - beta)*a_0)$
- Rotational RHS built the same way from angular momentum and torque
- Info printouts of matrix size and of input/output sixDOFx

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+57 -11	78, 80-82, 253-254, 266-290, 292-307, 325-326, 329-330, 332, 467-468, 471-472 (+1 more)

adding sixDOF_source

Replaces the hard-coded values with a 24-entry sixDOF_source vector (previous position, orientation, velocity, angular momentum, accelerations and torques). The beam model still fills it with dummy numbers. Also computes Newmark velocity and angular-momentum updates, which are printed but not used.

- coupledTotalLagNewtonRaphsonBeamEvolve.C: builds sixDOF_source(24,1) filled with placeholder constants
- BlockEigenSolverOF::solve gains a sixDOF_source argument and unpacks it
- Bug: the orientation entries are read into rb_angMomentum_prev, which is then overwritten
- rb_v and rb_angMomentum are computed and printed only

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+36 -2	540, 542–575, 579
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+93 -22	230–231, 274, -5 at 277, 279–338, 342, 344–363, 366, 368–369, 372–373, 377 (+5 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+2 -1	135–136

Amir- start coupling

Adds commented-out code in the beam evolve loop for looking up the fluid mesh (region0) from the beam's object registry. No functional change.

- coupledTotalLagNewtonRaphsonBeamEvolve.C: commented lookupObject<fvMesh>("region0")

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+4 -1	527–530

Add rigid-body coupling support to beam solver

Adds named structs that carry rigid-body data from the motion solver into the beam and back, in place of the raw arrays. The six appended rows are still an uncoupled identity block, and the result is only printed.

- New RigidBodyCouplingData.H: RigidBodyState, RigidBodyStepData, RigidBodySolution
- beamModel::setRigidBodyStepData (empty virtual), overridden in coupledTotalLagNewtonRaphsonBeam with rigidBodyData_ and rigidBodyDataValid_
- BlockEigenSolverOF::solve(solVec, source, RigidBodySolution&, const RigidBodyStepData&); zero state used if no data is supplied
- Adds backup copies original_BlockEigenSolverOF.C/.H

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+4 -0	56, 715–717
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/RigidBodyCouplingData.H	+97 -0	1–97
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+6 -1	655–657, 660–662
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+22 -1	56, 208–219, 252, 381–388
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+35 -39	542–547, 549, 551–570, 572–579
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+231 -282	17–19, -7 at 26, 39–63, -4 at 64, 66–80, 83–84, -7 at 95, 97, -4 at 98, 100–101 (+80 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+15 -13	1, 36, 39, -2 at 47, -1 at 49, 51–53, -3 at 55, 59, 83, 92 (+8 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/original_BlockEigenSolverOF.C	+576 -0	1–576
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/original_BlockEigenSolverOF.H	+152 -0	1–152

Propagate rigid body state into fv beam coupling

Passes the body's previous and current state (displacement, orientation, velocity, angular momentum, acceleration, torque) to the beam before each beam solve. Bumps beamFoam to 6c25265.

- finiteVolumeBeam::restrain fills RigidBodyStepData and calls beam.setRigidBodyStepData
- sixDoFRigidBodyMotionFvBeam: new state0() accessor; initialCentreOfRotation() declaration moved
- Make/options adds the coupledTotalLagNewtonRaphsonBeam include path
- src/beamFoam -> 6c25265

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1
src/sDoFRGBFvBeam/Make/options	+2 -1	12–13
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.C	+40 -0	34, 266–304
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/sixDoFRigidBodyMotionFvBeam.H	+9 -4	82, 170, 244–247, 279–281
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/sixDoFRigidBodyMotionFvBeamI.H	+6 -0	230–235

beamFoam a94db8d 2026-04-30 Colm Mc Alister

working rigidBodyAndBeam case

Adds the rigidBodyAndBeam tutorial: an interFoam floating box tied to a beam through the moorFV sixDoFRigidBodyMotionFvBeam solver and finiteVolumeBeam restraint. Also adds debug printouts in the beam evolve loop and a build fix.

- tutorials/rigidBodyAndBeam: blockMesh/snappyHexMesh case, beam region, dynamicMeshDict with FvBeamNewmark solver and finiteVolumeBeam restraint
- Allwmake builds only createBeamMesh and setInitialPositionBeam utilities
- Make/options adds the coupledTotalLagNewtonRaphsonBeam include path
- coupledTotalLagNewtonRaphsonBeamEvolve.C: 'CTLNRB debug' Info lines

File	Lines	Changed line ranges (after the commit)
Allwmake	+5 -1	34–38
src/wireBunchingModels/Make/options	+1 -0	10
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+26 -0	59–66, 608–613, 617–621, 626–632
tutorials/rigidBodyAndBeam/Allclean	+19 -0	1–19
tutorials/rigidBodyAndBeam/Allrun	+10 -0	1–10
tutorials/rigidBodyAndBeam_short/Allclean	+16 -0	1–16
tutorials/rigidBodyAndBeam_short/Allrun	+10 -0	1–10
tutorials/rigidBodyAndBeam/ (case files)	+946623 -0	97 files
tutorials/rigidBodyAndBeam_short/ (case files)	+948569 -0	96 files

moorFV 1e5809f 2026-04-30 Colm Mc Alister

Update beamFoam rigid body beam test

Adds debug printouts of the attachment displacement, beam W values and attachment/anchor forces around the beam evolve call. Bumps beamFoam to a94db8d (the rigidBodyAndBeam tutorial).

- finiteVolumeBeam.C: 'finiteVolumeBeam debug' Info lines before and after evolve
- src/beamFoam -> a94db8d

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.C	+18 -0	266–272, 316–321, 338–342

beamFoam 7a41f13 2026-04-30 Colm Mc Alister

edit tutorial/rigidBodyAndBeam/Allrun

Changes the rigidBodyAndBeam Allrun to run interFoam after meshing rather than stopping after mesh generation.

- tutorials/rigidBodyAndBeam/Allrun: adds runApplication interFoam

File	Lines	Changed line ranges (after the commit)
tutorials/rigidBodyAndBeam/Allrun	+3 -3	7, 9–10

beamFoam ab56a8d 2026-05-01 Colm Mc Alister

rigidBodyAndBeam benchmark results

Commits a complete run of the rigidBodyAndBeam case (meshes, time directories, logs) as rigidBodyAndBeam_coupledSolver for reference. No source code changes; about 121 million lines of generated output.

- tutorials/rigidBodyAndBeam_coupledSolver: polyMesh, time directories, uniform/sixDoFRigidBodyMotionFvBeamState
- log.blockMesh, log.snappyHexMesh, log.interFoam

File	Lines	Changed line ranges (after the commit)
tutorials/rigidBodyAndBeam_coupledSolver/ (case files)	+121367244 -0	670 files

Log BlockEigen rigid-body motion history

Writes the rigid-body values from each BlockEigen solve to a postProcessing history file and uses the real time step in place of $\Delta t = 1$. Adds a comparison case and a plotting script.

- BlockEigenSolverOF.C: writeBlockEigenRigidBodyMotion writes
postProcessing/BlockEigenSolve_rigidBodyMotion/<startTime>/BlockEigenSolve_rigidBodyMotion.dat
- BlockEigenSolverOF::solve takes const Time& runTime; deltaT = runTime.deltaTValue()
- tutorials/rigidBodyAndBeam_compareSixDoF_Eigen case
- plot_rigid_body_displacement_comparison.py

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+2 -1	577–578
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+136 -2	28–30, 84–199, 357–358, 389, 642–645, 648–657
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+3 -1	52, 142–143
tutorials/rigidBodyAndBeam/plot_rigid_body_displacement_comparison.py	+279 -0	1–279
tutorials/rigidBodyAndBeam_compareSixDoF_Eigen/Allclean	+19 -0	1–19
tutorials/rigidBodyAndBeam_compareSixDoF_Eigen/Allrun	+10 -0	1–10
tutorials/rigidBodyAndBeam_compareSixDoF_Eigen/plot_rigid_body_displacement_comparison.py	+279 -0	1–279
tutorials/rigidBodyAndBeam_compareSixDoF_Eigen/ (case files)	+946623 -0	97 files

Make BlockEigen rigid body solution authoritative

Adds the beam-end reaction force and moment to the rigid-body RHS as an explicit term, and lets moorFV copy the BlockEigen rigid-body result into the motion state. The rigid-body rows are still an identity block, so the result is an explicit Newmark update, not an implicit solve.

- RigidBodyForceCoupling struct; RigidBodyStepData gains mass, momentOfInertia, centreOfRotation, attachmentPoint; RigidBodySolution gains velocity, angularMomentum, acceleration, torque
- beamProperties keywords blockEigenForceCoupling and blockEigenKinematicCoupling; dynamicMeshDict keyword blockEigenAuthoritativeMotion
- Kinematic option adds one-way beam-row -> rigid-body-column terms (Cw/pDelta, moment arm set to zero); beamModel::getRigidBodySolution
- Adds two Codex .md notes, plot_rigidBodyAndBeam_new_vs_old.py and a regression check script (Alltest)

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+5 -0	718–722
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/RigidBodyCouplingData.H	+48 -1	53, 82–97, 104–107, 112–138
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+1 -0	662
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+18 -0	220–225, 393–404
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+270 -45	41, -8 at 59, 182–316, 669–753, 756–784, -14 at 797, 810–811, -6 at 832, 836–853, -7 at 858
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolver0F/BlockEigenSolver0F.C	+222 -68	-31 at 54, 165–263, -1 at 280, -3 at 282, 335–377, 450–481, -2 at 501, 528–565, 579, 587 (+6 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolver0F/BlockEigenSolver0F.H	+30 -0	84–98, 141–155
tutorials/plot_rigidBodyAndBeam_new_vs_old.py	+423 -0	1–423
tutorials/rigidBodyAndBeam/Alltest	+3 -0	1–3
tutorials/rigidBodyAndBeam/check_block_eigen_authoritative_regression.py	+196 -0	1–196
BEAM_ATTACHMENT_KINEMATICS_OWNERSHIP.md	+104 -0	document
BLOCK_EIGEN_RIGID_BODY_COUPLING_PROGRESS.md	+361 -0	document
tutorials/rigidBodyAndBeam/ (case files)	+4 -0	2 files

Apply authoritative BlockEigen rigid body motion

With blockEigenAuthoritativeMotion on, copies the BlockEigen rigid-body result into the motion state and returns zero restraint force and moment. FvBeamNewmark then recomputes velocity, position and orientation after restrain(), so the coupling stays partitioned. Bumps beamFoam to c78419b.

- New sixDoFRigidBodyMotionFvBeam::applyBlockEigenSolution sets centreOfRotation, v, a, pi, tau and Q
- finiteVolumeBeam::restrain takes a non-const motion; sends mass, inertia, centre of rotation and attachment point; reads blockEigenAuthoritativeMotion
- sixDoFRigidBodyMotionFvBeamRestraint::restrain signature made non-const
- sDoFRGBState/Make/options adds BEAMFOAM_DIR include paths; src/beamFoam -> c78419b

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.C	+53 -23	225, -7 at 265, 273, 276, 279, 296-301, 314-321, 323-324, 326-351, 368-369 (+1 more)
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.H	+1 -1	132
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/sixDoFRigidBodyMotionFvBeamRestraint/sixDoFRigidBodyMotionFvBeamRestraint.H	+1 -1	140
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/sixDoFRigidBodyMotionFvBeam.C	+27 -0	229-255
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/sixDoFRigidBodyMotionFvBeam.H	+7 -0	57, 317-322
src/sDoFRGBState/Make/options	+3 -1	6-8

Add finiteVolumeBeam attachment force history output

Mainly adds rigid-body-row -> beam-column coefficients, so the appended block is two-way coupled when the kinematic option is on (the tutorial has it off). The message's 'attachment force history' change is only a renamed input file in the plot script.

- BlockEigenSolverOF: rigidBodyBeamForceWCoeff_, rigidBodyBeamForceThetaCoeff_, rigidBodyMomentArm_; localSpinTensor helper
- Reciprocal terms are force stiffnesses (Cw/pDelta, CQTheta) placed in displacement rows without a dt^2/m scale
- Moment arm now attachmentPoint - centreOfRotation
- Adds BLOCK_EIGEN_SOLVER_COUPLING_CHANGES.md; plot script reads attachmentForcebeam.dat

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+13 -1	217-219, 271-273, 275, 279-281, 781-783
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+93 -17	-6 at 48, 259-275, 328, 354-360, 385-424, 433, 514-516, 531-533, 546-548, 564-568 (+4 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+13 -1	96-104, 150, 161-163
tutorials/plot_rigidBodyAndBeam_new_vs_old.py	+38 -9	56, 362, 370, 377, 391-417, 427, 434, 445, 449-452
BLOCK_EIGEN_SOLVER_COUPLING_CHANGES.md	+165 -0	document

Update 10/8/2026

Adds the beamAttachmentKinematics keyword. With solvedByBlockEigen the restraint stops fixing the beam-end W and leaves it to beamFoam's staged mode. Also writes an attachment-force history file. No submodule bump.

- finiteVolumeBeam: beamAttachmentKinematics_ (prescribedByRigidBody default, or solvedByBlockEigen), stored in the restraint state
- Sets rbData.solveAttachmentKinematicsInBlockEigen
- attachmentForceFilePtr_ writes attachmentForce<name>.dat
- Pairs with beamFoam 6bc4d04

File	Lines	Changed line ranges (after the commit)
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.C	+59 -2	88, 96, 164–170, 200–208, 225, 282–299, 336–337, 436–444, 468–473, 485–489
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.H	+4 -0	74–76, 93

Update 10/8/2026

Adds blockEigenKinematicCouplingMode (staged or fullyImplicit). Staged, the tutorial default, sets the beam end to the body's current displacement as a fixed value and adds no off-diagonal terms, so the rigid-body rows are an uncoupled identity block. Only fullyImplicit adds coupling terms.

- beamProperties: blockEigenKinematicCouplingMode (default staged); dynamicMeshDict: beamAttachmentKinematics solvedByBlockEigen
- coupledTotalLagNewtonRaphsonBeamBoundaryConditions.C: end-face source uses blockEigenStagedAttachmentDisplacement_ in staged mode, no source in fullyImplicit mode
- BlockEigenSolverOF: rigid-body unknowns become increments about rigidBodyAttachmentDisplacementPrevious_; coupling block only when rigidBodyKinematicCoupling_ (fullyImplicit)
- RigidBodyStepData::solveAttachmentKinematicsInBlockEigen; adds two Codex development logs

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/RigidBodyCouplingData.H	+3 -1	87, 97–98
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+3 -0	662–664
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+9 -0	220–228
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamBoundaryConditions.C	+44 -11	386–430
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+135 -43	–9 at 145, –4 at 168, 201–313, 320, 325, –6 at 341, 347–348, 350, 371–372, 848 (+7 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+71 -47	346, –8 at 354, 378–379, 381–382, 384–388, 390–392, 394–398, 400–437, 446–447, 531 (+6 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+5 -0	105–107, 161, 168
tutorials/plot_rigidBodyAndBeam_new_vs_old.py	+62 -42	25–58, 67–76, 192–197, 201–205, 353, 400–401, 431–432, 471–472
rigidBodyAndBeamCoupledSolverDevelopmentLog.txt	+541 -0	document
rigidBodyKinematicCouplingDevelopment.txt	+217 -0	document
tutorials/rigidBodyAndBeam/ (case files)	+3 -1	2 files

Attachment Coefficient changed to approximation of Hooke's Law

Scales the reciprocal rigid-body-row coefficients by $dt^2 \beta / m$ so they act like a Newmark-scaled attachment stiffness. This code only runs in fullyImplicit mode, which none of the tutorial cases use. Adds three more tutorial cases.

- BlockEigenSolverOF: rigidBodyNewmarkDisplacementScale = $\text{sqr}(\text{deltaT}) * \text{rbNewmarkBeta} / \text{mass}$ passed into convertFoamMatrixToEigenMatrix
- Rigid-body rows get +scale*K on rigid-body columns and -scale*K on beam columns; torque terms and localSpinTensor removed
- New cases rigidBodyAndBeam_ForceCoupledSolver, _fullTime, _shortTime (staged or force-only)
- Adds rigidBodyImplicitAttachmentCoefficientChanges.txt

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+37 -38	-16 at 259, 272, 371-377, 379-381, 387-390, 396-397, 406-407, 415, 417, 573-587 (+1 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+1 -0	131
tutorials/plot_rigidBodyAndBeam_new_vs_old.py	+2 -2	26, 106
tutorials/rigidBodyAndBeam_ForceCoupledSolver/Allclean	+19 -0	1-19
tutorials/rigidBodyAndBeam_ForceCoupledSolver/Allrun	+10 -0	1-10
tutorials/rigidBodyAndBeam_ForceCoupledSolver/Alltest	+3 -0	1-3
tutorials/rigidBodyAndBeam_ForceCoupledSolver/check_block_eigen_authoritative_regression.py	+196 -0	1-196
tutorials/rigidBodyAndBeam_ForceCoupledSolver/plot_rigidBody_displacement_comparison.py	+279 -0	1-279
tutorials/rigidBodyAndBeam_fullTime/Allclean	+19 -0	1-19
tutorials/rigidBodyAndBeam_fullTime/Allrun	+10 -0	1-10
tutorials/rigidBodyAndBeam_fullTime/Alltest	+3 -0	1-3
tutorials/rigidBodyAndBeam_fullTime/check_block_eigen_authoritative_regression.py	+196 -0	1-196
tutorials/rigidBodyAndBeam_fullTime/plot_rigidBody_displacement_comparison.py	+279 -0	1-279
tutorials/rigidBodyAndBeam_shortTime/Allclean	+19 -0	1-19
tutorials/rigidBodyAndBeam_shortTime/Allrun	+10 -0	1-10
tutorials/rigidBodyAndBeam_shortTime/Alltest	+3 -0	1-3
tutorials/rigidBodyAndBeam_shortTime/check_block_eigen_authoritative_regression.py	+196 -0	1-196
tutorials/rigidBodyAndBeam_shortTime/plot_rigidBody_displacement_comparison.py	+279 -0	1-279
rigidBodyImplicitAttachmentCoefficientChanges.txt	+181 -0	document
tutorials/rigidBodyAndBeam_ForceCoupledSolver/ (case files)	+946627 -0	97 files
tutorials/rigidBodyAndBeam_fullTime/ (case files)	+946629 -0	97 files
tutorials/rigidBodyAndBeam_shortTime/ (case files)	+946629 -0	97 files

Add efficiency plotting script

Adds a Python script that parses two interFoam logs and compares run time, PIMPLE iterations and beam Newton iterations between the new and original coupling cases.

- tutorials/plot_rigidBodyAndBeam_algorithm_efficiency_new_vs_old.py (compares rigidBodyAndBeam_fullTime with rigidBodyAndBeam_compareSixDoF_Eigen)

File	Lines	Changed line ranges (after the commit)
tutorials/plot_rigidBodyAndBeam_algorithm_efficiency_new_vs_old.py	+749 -0	1-749

beamFoam 778e43d 2026-09-30 Colm Mc Alister

Add Newton iteration interval plots to efficiency script

Adds plots to the efficiency script: Newton iterations per time step, summed over fixed time intervals, and cumulative.

- plot_rigidBodyAndBeam_algorithm_efficiency_new_vs_old.py: interval and cumulative Newton-iteration plots

File	Lines	Changed line ranges (after the commit)
tutorials/plot_rigidBodyAndBeam_algorithm_efficiency_new_vs_old.py	+165 -7	58–59, 524–573, 639, 650–758, 874–876

2.4 30 September – 1 October 2026: review and monolithic coupling, Phases 0–4 (Colm Mc Alister, with Claude)

Review of the earlier path, the development plan, then beamFoam solving the body monolithically (rigidBodyEnd), moorFV's beamFoamCoupled solver, and validation in interFoam.

moorFV 8b99fd5 2026-09-30 Colm Mc Alister

Update beamFoam submodule to latest colm_6DOF_solver

Moves the beamFoam submodule pointer from c78419b to 778e43d, picking up de010c8 through 778e43d (staged/fullyImplicit modes, coefficient scaling, efficiency scripts).

- src/beamFoam: c78419b -> 778e43d

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1

moorFV 74005ba 2026-09-30 Colm Mc Alister

Use BEAMFOAM_DIR for beamFoam include paths in sDoFRGBFvBeam

Replaces the hard-coded /Users/... and WM_PROJECT_USER_DIR beamFoam include paths in sDoFRGBFvBeam with \$(BEAMFOAM_DIR), as sDoFRGBState and CI already use.

- src/sDoFRGBFvBeam/Make/options

File	Lines	Changed line ranges (after the commit)
src/sDoFRGBFvBeam/Make/options	+2 -2	12–13

beamFoam bda04a6 2026-09-30 Colm Mc Alister

Move Codex development notes into codexLogs and add PR summary

Moves the six Codex development notes from the repository root into codexLogs/ and adds a PR summary PDF. The PDF described the work as monolithic coupling; f84a03a later corrects it.

- codexLogs/: the three .md notes and three .txt logs (renames only)
- Adds codexLogs/rigidBodyCouplingPRSummary.pdf

File	Lines	Changed line ranges (after the commit)
BEAM_ATTACHMENT_KINEMATICS_OWNERSHIP.md => codexLogs/BEAM_ATTACHMENT_KINEMATICS_OWNERSHIP.md	+0 -0	document
BLOCK_EIGEN_RIGID_BODY_COUPLING_PROGRESS.md => codexLogs/BLOCK_EIGEN_RIGID_BODY_COUPLING_PROGRESS.md	+0 -0	document
BLOCK_EIGEN_SOLVER_COUPLING_CHANGES.md => codexLogs/BLOCK_EIGEN_SOLVER_COUPLING_CHANGES.md	+0 -0	document
rigidBodyAndBeamCoupledSolverDevelopmentLog.txt => codexLogs/rigidBodyAndBeamCoupledSolverDevelopmentLog.txt	+0 -0	document
codexLogs/rigidBodyCouplingPRSummary.pdf	bin	document
rigidBodyImplicitAttachmentCoefficientChanges.txt => codexLogs/rigidBodyImplicitAttachmentCoefficientChanges.txt	+0 -0	document
rigidBodyKinematicCouplingDevelopment.txt => codexLogs/rigidBodyKinematicCouplingDevelopment.txt	+0 -0	document

Add monolithic and loop rigidBodyAndBeam cases; group older cases

Adds two comparison cases and groups the older cases under tutorials/rigidBodyAndBeam/. The 'monolithic' case uses staged mode, so it is still a partitioned scheme with an identity rigid-body block; the 'loop' case turns all BlockEigen coupling switches off.

- tutorials/rigidBodyAndBeam_monolithic: force coupling, staged kinematic coupling, solvedByBlockEigen, blockEigenAuthoritativeMotion
- tutorials/rigidBodyAndBeam_loop: original partitioned coupling via sixDoFRigidBodyMotionFvBeam
- Moves rigidBodyAndBeam, _compareSixDoF_Eigen, _coupledSolver, _ForceCoupledSolver, _fullTime, _short, _shortTime into tutorials/rigidBodyAndBeam/
- Both plotting scripts now point at the monolithic and loop cases

File	Lines	Changed line ranges (after the commit)
tutorials/plot_rigidBodyAndBeam_algorithm_efficiency_new_vs_old.py	+2 -2	27-28
tutorials/plot_rigidBodyAndBeam_new_vs_old.py	+2 -2	25-26
tutorials/rigidBodyAndBeam/{ => rigidBodyAndBeam}/Allclean	+0 -0	1-19, 1-19
tutorials/rigidBodyAndBeam/{ => rigidBodyAndBeam}/Allrun	+0 -0	1-10, 1-10
tutorials/rigidBodyAndBeam/{ => rigidBodyAndBeam}/Alltest	+0 -0	1-3
tutorials/rigidBodyAndBeam/{ => rigidBodyAndBeam}/check_block_eigen_authoritative_regression.py	+0 -0	1-196
tutorials/rigidBodyAndBeam/{ => rigidBodyAndBeam}/plot_rigid_body_displacement_comparison.py	+0 -0	1-279, 1-279
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_ForceCoupledSolver/Allclean	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_ForceCoupledSolver/Allrun	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_ForceCoupledSolver/Alltest	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_ForceCoupledSolver/check_block_eigen_authoritative_regression.py	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_ForceCoupledSolver/plot_rigid_body_displacement_comparison.py	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_compareSixDoF_Eigen/Allclean	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_compareSixDoF_Eigen/Allrun	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_compareSixDoF_Eigen/plot_rigid_body_displacement_comparison.py	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_fullTime/Allclean	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_fullTime/Allrun	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_fullTime/Alltest	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_fullTime/check_block_eigen_authoritative_regression.py	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_fullTime/plot_rigid_body_displacement_comparison.py	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_short/Allclean	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_short/Allrun	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_shortTime/Allclean	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_shortTime/Allrun	+0 -0	rename only

tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_shortTime/Alltest	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_shortTime/check_block_eigen_authoritative_regression.py	+0 -0	rename only
tutorials/{ => rigidBodyAndBeam}/rigidBodyAndBeam_shortTime/plot_rigid_body_displacement_comparison.py	+0 -0	rename only
tutorials/rigidBodyAndBeam_loop/Allclean	+19 -0	
tutorials/rigidBodyAndBeam_loop/Allrun	+10 -0	
tutorials/rigidBodyAndBeam_loop/plot_rigid_body_displacement_comparison.py	+279 -0	
... 5 more code files in tutorials/rigidBodyAndBeam_monolithic/Allclean (1), tutorials/rigidBodyAndBeam_monolithic/Allrun (1), tutorials/rigidBodyAndBeam_monolithic/Alltest (1), tutorials/rigidBodyAndBeam_monolithic/check_block_eigen_authoritative_regression.py (1), tutorials/rigidBodyAndBeam_monolithic/plot_rigid_body_displacement_comparison.py (1)		
tutorials/rigidBodyAndBeam/rigidBodyAndBeam/ (case files)	+0 -0	97 files
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_ForceCoupledSolver/ (case files)	+0 -0	97 files
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_compareSixDoF_Eigen/ (case files)	+0 -0	97 files
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_coupledSolver/ (case files)	+0 -0	670 files
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_fullTime/ (case files)	+0 -0	97 files
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_short/ (case files)	+0 -0	96 files
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_shortTime/ (case files)	+0 -0	97 files
tutorials/rigidBodyAndBeam_loop/ (case files)	+946624 -0	97 files
tutorials/rigidBodyAndBeam_monolithic/ (case files)	+946630 -0	97 files

moorFV **f024a62** 2026-09-30 Colm Mc Alister

Update beamFoam submodule to latest colm_6DOF_solver

Moves the beamFoam submodule pointer to fb5d6c1, which adds codexLogs/ and the rigidBodyAndBeam_monolithic and rigidBodyAndBeam_loop cases.

- src/beamFoam: 778e43d -> fb5d6c1 (includes bda04a6)

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1

beamFoam **f84a03a** 2026-09-30 Colm Mc Alister

Correct PR summary PDF to match what the coupling code does

Replaces the PR summary PDF with a version checked against the code. It states that staged-mode rigid-body rows are an uncoupled identity block and that FvBeamNewmark recomputes the body state, and swaps in the monolithic-vs-loop benchmark.

- codexLogs/rigidBodyCouplingPRSummary.pdf (binary update)

File	Lines	Changed line ranges (after the commit)
codexLogs/rigidBodyCouplingPRSummary.pdf	bin	document

moorFV 2dc4bfd 2026-09-30 Colm Mc Alister

Update beamFoam submodule to latest colm_6DOF_solver

Moves the beamFoam submodule pointer to f84a03a, the corrected PR summary PDF.

- src/beamFoam: fb5d6c1 -> f84a03a

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1

beamFoam 4ab6dc8 2026-09-30 Colm Mc Alister

Add monolithic coupling development plan PDF

Adds a PDF plan for a real monolithic beam / rigid-body coupling, which the later Phase 0-4 commits follow.

- codexLogs/monolithicCouplingPlan.pdf (new binary)

File	Lines	Changed line ranges (after the commit)
codexLogs/monolithicCouplingPlan.pdf	bin	document

Add rigidBodyEnd and standalone beam / rigid-body tests (Phase 0)

Phase 0. Adds rigidBodyEnd, a point-mass body owned by beamFoam and attached to a beam end. Also adds partitioned coupling with converged coupling iterations, and standalone tests against analytic results. Fixes two existing bugs.

- rigidBodyEnd.H/.C (new): mass, external force, gravity, Newmark update, newTimeStep() by time index, history file postProcessing/rigidBodyEnd
- coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C (new): reads rigidBodyCoupling none | partitioned | monolithic; evolve() dispatches to evolvePartitionedRigidBodyEnd(), and the old evolve() becomes evolveBeam()
- Bug fix, Evolve.C: the Newmark predictor is applied once per time step (newmarkPredictorTimeIndex_), not on every repeated solve
- Bug fix, beamModel.C: the gravity rho check moved to after rho_ is filled (it crashed cases with a g file and per-beam rho); new tutorials/rigidBodyEnd (hangingMass, axialOscillation, pendulum) with checkRigidBodyEnd.py

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/Make/files	+2 -0	14–15
src/wireBunchingModels/beamModels/beamModel/beamModel.C	+10 -14	368–369, 549–556
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+3 -0	666, 1116–1117
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+27 -0	45, 58, 237–249, 367–378
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+9 -2	56, 148–153, 155–156
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C	+280 -0	1–280
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.C	+215 -0	1–215
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.H	+273 -0	1–273
tutorials/rigidBodyEnd/Allclean	+7 -0	1–7
tutorials/rigidBodyEnd/Allrun	+11 -0	1–11
tutorials/rigidBodyEnd/axialOscillation/Allclean	+7 -0	1–7
tutorials/rigidBodyEnd/axialOscillation/Allrun	+6 -0	1–6
tutorials/rigidBodyEnd/checkRigidBodyEnd.py	+218 -0	1–218
tutorials/rigidBodyEnd/hangingMass/Allclean	+7 -0	1–7
tutorials/rigidBodyEnd/hangingMass/Allrun	+6 -0	1–6
tutorials/rigidBodyEnd/pendulum/Allclean	+7 -0	1–7
tutorials/rigidBodyEnd/pendulum/Allrun	+6 -0	1–6
tutorials/rigidBodyEnd/README.md	+43 -0	document
tutorials/rigidBodyEnd/axialOscillation/ (case files)	+361 -0	10 files
tutorials/rigidBodyEnd/hangingMass/ (case files)	+361 -0	10 files
tutorials/rigidBodyEnd/pendulum/ (case files)	+361 -0	10 files

Update beamFoam submodule to latest colm_6DOF_solver

Moves the beamFoam submodule pointer to a2dcb75 (Phase 0: rigidBodyEnd and standalone beam / rigid-body tests), which also includes the 4ab6dc8 plan PDF.

- src/beamFoam: f84a03a -> a2dcb75

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1

beamFoam 0b3fa2b 2026-09-30 Colm Mc Alister

Mark Phase 0 completed in monolithic coupling plan

Marks Phase 0 as completed in the development plan PDF.

File	Lines	Changed line ranges (after the commit)
codexLogs/monolithicCouplingPlan.pdf	bin	document

beamFoam 359e651 2026-09-30 Colm Mc Alister

Add monolithic beam / rigid-body translation coupling (Phase 1)

Phase 1. Solves the body's translation in the same Newton iteration as the beam, inside BlockEigenSolverOF: true monolithic coupling for translation.

- RigidBodyCouplingData.H: new RigidBodyMonolithicCoupling (attachment cell, beam-row and body-row coefficients, body source)
- BlockEigenSolverOF.C/.H: setMonolithicCoupling(); body columns +CQW/pDelta on the end cell's force rows and +CMQW/pDelta on its moment rows; body rows $R = m a + c v - F_{ext} - m g + Q_b$; the increment is returned in rigidBodySolution
- RigidBodyEnd.C: evolveMonolithicRigidBodyEnd(), setRigidBodyEndIterate(), rigidBodyEndMonolithicCoupling() (boundary terms assembled alone into zeroed fields), checkRigidBodyEndJacobian() (finite-difference check)
- Evolve.C: hooks before updateCoeffs and after the solve ($x += dx$; the attachment W is updated); deltaXNorm includes the body increment

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/RigidBodyCouplingData.H	+42 -0	122–163
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+1 -0	667
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+32 -0	244–246, 382–410
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+40 -1	263–279, 905–912, 991–1003, 1009–1010
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C	+245 -12	31–33, -8 at 119, 169, 187–191, 205, 245, 271–504
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.C	+16 -4	91, 174–187, -3 at 196, 198
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.H	+18 -0	50–51, 113–115, 230–234, 258–265
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.C	+77 -0	331–398, 782–790
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/BlockEigenSolverOF.H	+11 -0	111–113, 186–193
tutorials/rigidBodyEnd/Allclean	+2 -0	8–9
tutorials/rigidBodyEnd/Allrun	+17 -2	2–3, 8, 10–23
tutorials/rigidBodyEnd/checkRigidBodyEnd.py	+93 -10	3–5, 122, 138, 171, 208–292, 294, 297
codexLogs/monolithicCouplingPlan.pdf	bin	document
tutorials/rigidBodyEnd/README.md	+24 -13	document
tutorials/rigidBodyEnd/ (case files)	+1 -0	1 file
tutorials/rigidBodyEnd/axialOscillation/ (case files)	+4 -0	1 file
tutorials/rigidBodyEnd/pendulum/ (case files)	+4 -0	1 file

moorFV 70e8053 2026-09-30 Colm Mc Alister

Update beamFoam submodule to latest colm_6DOF_solver

Moves the beamFoam submodule pointer to 359e651 (Phase 1: monolithic beam / rigid-body translation coupling).

- src/beamFoam: a2dcb75 -> 359e651

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1

Add monolithic rigid-body rotation coupling (Phase 2)

Phase 2. Adds body rotation to both couplings: inertia tensor, centre of mass away from the attachment, angular velocity, damping and external moment, all solved monolithically with the beam.

- rigidBodyEnd: orientation $\exp(\theta) \cdot Q_n$ with Newmark on θ ; moment balance $J \alpha + \omega \times J \omega + c_r \omega - M_{ext} - r \times F_{beam}$; rotationalResidual(), rotationJacobian() with the SO(3) left Jacobian $T(\theta)$; arm(), attachmentDisplacement()
- BlockEigenSolverOFC: rotation rows and columns, plus translation-rotation cross terms
- RigidBodyCouplingData.H: rotation coefficients (bodyRot*, beam*RowRotCoeff) and RigidBodyEndState
- New tutorials hangingBody, freeRotation and compoundPendulum; checkRigidBodyEnd.py grows to 54 checks

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/RigidBodyCouplingData.H	+34 -5	122–125, 130–131, 153–170, 174, 181–189
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.C	+2 -0	668–669
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+16 -7	247–251, 399–409
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamEvolve.C	+21 -7	277, 992–994, 998–1007, 1010–1015, 1024
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C	+165 -69	171–174, 211–213, 217, 225, 227–228, 230, 233–238, 245, 253–255, 257 (+37 more)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.C	+368 -8	38–103, 129–132, 137–180, 198–201, 203–208, 217, 222–228, 230, 250–277, 281–286 (+10 more)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.H	+144 -12	30–39, 48–52, 54–56, 60–61, 67–68, 84, 110, 113–124, 126–130, 157–169 (+6 more)
src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOFC/BlockEigenSolverOFC.C	+74 -1	395–464, 857–860
tutorials/rigidBodyEnd/Allclean	+1 -1	4
tutorials/rigidBodyEnd/Allrun	+1 -1	6
tutorials/rigidBodyEnd/checkRigidBodyEnd.py	+180 -7	60–99, 248–367, 377–381, 383, 405, 409–412, 433–435, 451, 461–464, 467
tutorials/rigidBodyEnd/compoundPendulum/Allclean	+7 -0	1–7
tutorials/rigidBodyEnd/compoundPendulum/Allrun	+6 -0	1–6
tutorials/rigidBodyEnd/freeRotation/Allclean	+7 -0	1–7
tutorials/rigidBodyEnd/freeRotation/Allrun	+6 -0	1–6
tutorials/rigidBodyEnd/hangingBody/Allclean	+7 -0	1–7
tutorials/rigidBodyEnd/hangingBody/Allrun	+6 -0	1–6
codexLogs/monolithicCouplingPlan.pdf	bin	document
tutorials/rigidBodyEnd/README.md	+20 -10	document
tutorials/rigidBodyEnd/compoundPendulum/ (case files)	+372 -0	10 files
tutorials/rigidBodyEnd/freeRotation/ (case files)	+369 -0	10 files
tutorials/rigidBodyEnd/hangingBody/ (case files)	+374 -0	10 files

Update beamFoam submodule to latest colm_6DOF_solver

Moves the beamFoam submodule pointer to 03bbd7d (Phase 2: monolithic rigid-body rotation coupling).

- src/beamFoam: 359e651 -> 03bbd7d

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1

Add beamModel interface for externally driven rigidBodyEnd (Phase 3)

Phase 3, beamFoam side. Adds a beamModel interface so a caller (moorFV) can create the body, set its applied load each step and read back the solved state. Adds the interFoam case using moorFV's new beamFoamCoupled solver.

- beamModel.H: virtual initialiseRigidBodyEnd(dict, gravity), setRigidBodyEndExternalLoad(F, M) and getRigidBodyEndState()
- rigidBodyEnd: initial orientation; inertia and arm in body axes; setExternalLoad(); the history folder is named after the start time
- RigidBodyEnd.C: createRigidBodyEnd(); the Jacobian check uses central differences
- tutorials/rigidBodyAndBeam_beamFoamCoupled with checkCoupledState.py and Alltest (moorFV's state equals beamFoam's to about 2e-10)

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/beamModel/beamModel.H	+33 -0	718–750
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/RigidBodyCouplingData.H	+29 -0	193–221
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeam.H	+29 -0	383–393, 484–501
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C	+112 -30	79, 112–113, 116–137, 145, 180, 182–252, 592–593, 610–616, 620–624
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.C	+10 -5	113–114, 117–118, 205, 226, 229, 273, 311, 419
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/rigidBodyEnd.H	+29 -2	48, 50–51, 123, 125, 288–311
tutorials/rigidBodyAndBeam_beamFoamCoupled/Allclean	+19 -0	1–19
tutorials/rigidBodyAndBeam_beamFoamCoupled/Allrun	+10 -0	1–10
tutorials/rigidBodyAndBeam_beamFoamCoupled/Alltest	+3 -0	1–3
tutorials/rigidBodyAndBeam_beamFoamCoupled/checkCoupledState.py	+108 -0	1–108
tutorials/rigidBodyAndBeam_beamFoamCoupled/plot_rigid_body_displacement_comparison.py	+279 -0	1–279
tutorials/rigidBodyEnd/checkRigidBodyEnd.py	+4 -6	405, 408–410
codexLogs/monolithicCouplingPlan.pdf	bin	document
tutorials/rigidBodyAndBeam_beamFoamCoupled/ (case files)	+946627 -0	97 files

Add beamFoamCoupled sixDoF solver (Phase 3)

Phase 3, moorFV side. New sixDoF solver beamFoamCoupled: moorFV hands the fluid load to beamFoam, which solves beam and body together; moorFV then copies the solved state as it is.

- beamFoamCoupled.H/.C (new): applied load = fluid force/moment with weight + other restraints (explicit); calls finiteVolumeBeam::solveMonolithic; sets centre of rotation, Q , v , a , $\pi = J \cdot (Q^T \omega)$, τ
- finiteVolumeBeam.C/.H: solveMonolithic(motion, force, moment, solverDict); on the first call it sets up beamFoam's body from the motion (no gravity in beamFoam)
- sixDoFFvBeamSolver.H / I.H: protected accessors restraints(), constraints() and initialCentreOfRotation()
- Make/files registers the solver; the beamFoam submodule points to 83acbbb

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1
src/sDoFRGBFvBeam/Make/files	+1 -0	32
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers/beamFoamCoupled/beamFoamCoupled.C	+147 -0	1–147
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers/beamFoamCoupled/beamFoamCoupled.H	+136 -0	1–136
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers/sixDoFFvBeamSolver/sixDoFFvBeamSolver.H	+9 -0	91–99
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers/sixDoFFvBeamSolver/sixDoFFvBeamSolverI.H	+21 -0	66–86
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.C	+122 -1	97–98, 459–578
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.H	+18 -0	96–99, 146–159

Add Phase 4 validation: converged references, stress tests, docs

Renames codexLogs/ to documentation/. Its message describes all of Phase 4, but a failed git add left only the rename in this commit; the remaining files are in 74ebab7.

File	Lines	Changed line ranges (after the commit)
{codexLogs => documentation}/BEAM_ATTACHMENT_KINEMATICS_OWNERSHIP.md	+0 -0	document
{codexLogs => documentation}/BLOCK_EIGEN_RIGID_BODY_COUPLING_PROGRESS.md	+0 -0	document
{codexLogs => documentation}/BLOCK_EIGEN_SOLVER_COUPLING_CHANGES.md	+0 -0	document
{codexLogs => documentation}/monolithicCouplingPlan.pdf	bin	document
{codexLogs => documentation}/rigidBodyAndBeamCoupledSolverDevelopmentLog.txt	+0 -0	document
{codexLogs => documentation}/rigidBodyCouplingPRSummary.pdf	bin	document
{codexLogs => documentation}/rigidBodyImplicitAttachmentCoefficientChanges.txt	+0 -0	document
{codexLogs => documentation}/rigidBodyKinematicCouplingDevelopment.txt	+0 -0	document

Add the Phase 4 files missed from bf5bbbe

Phase 4. Adds the converged-reference interFoam cases, the stress-test generator, updated plotting scripts and the outer-corrector / Newmark note, and marks Phase 4 completed in the plan.

- rigidBodyAndBeam_loop_nOuter8, rigidBodyAndBeam_beamFoamCoupled_nOuter8: nOuterCorrectors 8, moveMeshOuterCorrectors yes
- rigidBodyAndBeam_stress: makeCases (stiffLine, lightBody, stiffLight, stiffLargeDt x 4 methods; removes 0/beam/CQ and CM so E takes effect), Allrun (parallel), summarise.py
- plot_rigidBodyAndBeam_new_vs_old.py rewritten for several cases with a reference and CASE_SET; the efficiency script gets stress-case labels and summed Newton iterations
- documentation/outerCorrectorsAndNewmark.pdf and documentation/source/ (HTML, model.py, figs.py); codexLogs references updated

File	Lines	Changed line ranges (after the commit)
src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C	+1 -1	33
tutorials/plot_rigidBodyAndBeam_algorithm_efficiency_new_vs_old.py	+53 -17	4-6, 9-11, 30-36, 38-56, 72-73, 173-174, 177-181, 468-470, 475-477, 491-493 (+1 more)
tutorials/plot_rigidBodyAndBeam_new_vs_old.py	+210 -341	4, 6-16, -1 at 18, 28, 33-55, 57-58, -11 at 64, 69, 76-77, -3 at 78 (+51 more)
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/Allclean	+19 -0	1-19
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/Allrun	+10 -0	1-10
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/Alltest	+3 -0	1-3
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/checkCoupledState.py	+108 -0	1-108
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/plot_rigidBody_displacement_comparison.py	+279 -0	1-279
tutorials/rigidBodyAndBeam_loop_nOuter8/Allclean	+19 -0	1-19
tutorials/rigidBodyAndBeam_loop_nOuter8/Allrun	+10 -0	1-10
tutorials/rigidBodyAndBeam_loop_nOuter8/plot_rigidBody_displacement_comparison.py	+279 -0	1-279
tutorials/rigidBodyAndBeam_stress/Allclean	+4 -0	1-4
tutorials/rigidBodyAndBeam_stress/Allrun	+38 -0	1-38
tutorials/rigidBodyAndBeam_stress/makeCases	+121 -0	1-121
tutorials/rigidBodyAndBeam_stress/summarise.py	+149 -0	1-149
documentation/monolithicCouplingPlan.pdf	bin	document
documentation/outerCorrectorsAndNewmark.pdf	bin	document
documentation/source/README.md	+15 -0	document
documentation/source/figs.py	+36 -0	document
documentation/source/growth.png	bin	document
documentation/source/model.py	+43 -0	document
documentation/source/monolithicCouplingPlan.html	+319 -0	document
documentation/source/newton.png	bin	document
documentation/source/order.py	+13 -0	document
documentation/source/outerCorrectorsAndNewmark.html	+243 -0	document
tutorials/rigidBodyEnd/README.md	+1 -1	document
tutorials/rigidBodyAndBeam_beamFoamCoupled/ (case files)	+3 -2	1 file
tutorials/rigidBodyAndBeam_beamFoamCoupled_nOuter8/ (case files)	+946634 -0	97 files
tutorials/rigidBodyAndBeam_loop/ (case files)	+3 -2	1 file
tutorials/rigidBodyAndBeam_loop_nOuter8/ (case files)	+946631 -0	97 files
tutorials/rigidBodyAndBeam_stress/ (case files)	+7 -0	1 file

Update beamFoam: Phase 4 validation and documentation

Points the beamFoam submodule at 74ebab7 (Phase 4 validation and documentation).

File	Lines	Changed line ranges (after the commit)
src/beamFoam	+1 -1	1

3. moorFV: net changes

All moorFV changes, branch point to now (the beamFoam submodule pointer is listed as `src/beamFoam`):

File	Lines	Type
src/beamFoam		
src/beamFoam	+1 -1	code
src/sDoFRGBFvBeam/Make		
src/sDoFRGBFvBeam/Make/files	+1 -0	code
src/sDoFRGBFvBeam/Make/options	+3 -1	code
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers		
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers/beamFoamCoupled/beamFoamCoupled.C	+147 -0	code
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers/beamFoamCoupled/beamFoamCoupled.H	+136 -0	code
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers/sixDoFFvBeamSolver/sixDoFFvBeamSolver.H	+9 -0	code
src/sDoFRGBFvBeam/sixDoFFvBeamSolvers/sixDoFFvBeamSolver/sixDoFFvBeamSolverI.H	+21 -0	code
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam		
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.C	+302 -7	code
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/finiteVolumeBeamRestraint/finiteVolumeBeam.H	+24 -1	code
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/restraints/sixDoFRigidBodyMotionFvBeamRestraint/sixDoFRigidBodyMotionFvBeamRestraint.H	+1 -1	code
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/sixDoFRigidBodyMotionFvBeam.C	+27 -0	code
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/sixDoFRigidBodyMotionFvBeam.H	+16 -4	code
src/sDoFRGBFvBeam/sixDoFRigidBodyMotionFvBeam/sixDoFRigidBodyMotionFvBeamI.H	+6 -0	code
src/sDoFRGBState/Make		
src/sDoFRGBState/Make/options	+3 -1	code

- **New sixDoF solver** `beamFoamCoupled` (`sixDoFFvBeamSolvers/beamFoamCoupled/` , 36393fb). The fluid force and moment (with the weight) and any other restraints form the applied load. `beamFoam` solves the beam and the body together, and the state it returns is copied into the motion without recomputation or relaxation.
- **finiteVolumeBeam restraint.**
 - `solveMonolithic()` (36393fb).
 - Earlier: `RigidBodyStepData` is passed to the beam, there are `beamAttachmentKinematics` and `blockEigenAuthoritativeMotion` options, and the attachment force has a history file (d72ed0a, 3dc0bb1, ffa51e3).
- **Interface change.** `sixDoFRigidBodyMotionFvBeamRestraint::restrain()` takes a non-const motion.
- **Base solver accessors:** `restraints()` , `constraints()` , `initialCentreOfRotation()` ; `sixDoFRigidBodyMotionFvBeam` gains `applyBlockEigenSolution()` and `state0()` .
- **Build.** `Make/options` include `beamFoam` headers through `$(BEAMFOAM_DIR)` (74005ba), replacing hard-coded paths.

- **Open points from the review:**

- The acceleration relaxation in `updateAcceleration` also applies on the first outer corrector of each step (`static bool first`).
- `main` has renamed `sDoFRGBFvBeam`, so the branch must be rebased.
- The `displacement*.dat` file holds the anchor displacement.

4. beamFoam merge commits

Merges bring branches together. Their content is already covered by the commits above.

Merge	Date	Author	Subject
1037fe2	2025-04-04	Seevani Bali	Merge branch 'openfoam-v2306' into openfoam-v2306-seevani
5340489	2025-07-24	Seevani Bali	Merge pull request #18 from solids4foam/openfoam-v2306-seevani
fdba2f7	2025-07-28	Seevani Bali	Merge pull request #20 from solids4foam/main
d50c3df	2025-08-07	Seevani Bali	Merge pull request #19 from solids4foam/openfoam-v2306-seevani
fdeb9d5	2025-08-12	Seevani Bali	Merge pull request #21 from solids4foam/pr/CTLNRB-solver-refactor
1e350b3	2025-08-22	Seevani Bali	Merge branch 'openfoam-v2306-seevani' into main
9987c88	2025-08-22	Seevani Bali	Merge pull request #24 from solids4foam/main
a77c8e3	2025-08-29	Seevani Bali	Merge pull request #25 from solids4foam/openfoam-v2306-seevani
aa40fd1	2025-11-18	Seevani Bali	Merge pull request #27 from solids4foam/feature-cloner-workflows
0ee0e78	2025-11-18	Seevani Bali	Merge pull request #28 from solids4foam/openfoam-v2306-seevani - Tutorial updates + Bug fix to inertia Euler terms
50ba069	2025-11-21	Seevani Bali	Merge pull request #29 from solids4foam/openfoam-v2306-seevani-momentumContribution
83042b6	2025-12-10	Philip Cardiff	Merge pull request #30 from solids4foam/openfoam-v2306-seevani

5. Open issues found while preparing this summary

Each of these was checked against the current code (beamFoam 74ebab7, moorFV f1bd195). None is fixed on the branch yet.

Where	Issue	From
beamModel.C:540–544	The "Missing E, G" <code>FatalErrorInFunction</code> ends with <code>endl</code> , not <code>abort(FatalError)</code> , so a case with no E or G carries on with zero stiffness.	f534931
beamModel.C:579–582	<code>startPatchName</code> and <code>endPatchName</code> are hard-coded to <code>left</code> and <code>right</code> ; any value in <code>beamProperties</code> is ignored.	cf11bd9
tutorials/rigidBodyAndBeam_*/0/beam/CQ, CM	Saved stiffness fields are read in preference to E and G (<code>READ_IF_PRESENT</code>), so editing E in <code>beamProperties</code> has no effect.	fb5d6c1 and earlier cases
moorFV sixDoFRigidBodyMotionFvBeam.C:312	<code>static bool first:</code> acceleration relaxation also	pre-existing

	applies on the first outer corrector of each step, mixing in the previous step's acceleration (a lag in <code>FvBeamNewmark</code>).	
tutorials/rigidBodyAndBeam/rigidBodyAndBeam_coupledSolver/ and other cases	Complete run output is committed: 670 files and 2.6 GB on disk for this case alone (about 121 million lines in ab56a8d). This makes up most of the PR diff.	ab56a8d
BlockEigenSolver0F/original_BlockEigenSolver0F.C/.H , editor #...# files	Backup and editor files are tracked.	6c25265 and earlier
BlockEigen staged / fullyImplicit path, blockEigen* switches; moorFV blockEigenAuthoritativeMotion	The earlier coupling path is still present alongside the monolithic one. It is a partitioned scheme, and <code>fullyImplicit</code> was never validated.	Feb – Aug 2026

Other mismatches between commit messages and diffs are noted on the commit cards above. Several are commits that don't compile on their own and are fixed in the next commit (e.g. e2272c1 and 95ad89d, fixed in f1bd715).

6. Phase 5: several mooring lines (plan)

The problem

- **Only one line is supported.** The monolithic coupling adds the body's six unknowns to *one* beam's BlockEigen system, and `beamFoamCoupled` stops if the body has a second `finiteVolumeBeam` restraint.
- **The target case has four.** `mooringCase-H12T20` has four lines (`beamone ... beamfour`). Each is a separate restraint with its own beam region and its own `beamModel` (`finiteVolumeBeam.C:65–70` , `beamModel::New` per restraint), so it has its own mesh and its own linear system. The body is a 3.144 kg floating box in waves. It runs with `FvBeamNewmark` , `accelerationRelaxation 0.7` and 3 outer correctors.

Two ways to couple several lines

Approach	How	Trade-offs
A. One combined matrix	Put every line and the body in one sparse system: body columns on each line's end cell, with the body rows summing all attachment forces. Easiest if all lines live in one <code>beamFoam</code> region, which <code>beamFoam</code> already supports for several beams via cell zones (<code>nBeams > 1</code> in the <code>CQ_ / CM_</code> set-up).	Reuses the Phase 1–2 code almost unchanged. But the case must be rebuilt with one beam region; the system grows with every line; it is serial; and all lines share one Newton loop.
B. Static condensation (recommended)	Each line keeps its own system. It is reduced to a 3×3 stiffness and a 3-vector residual at its attachment, and the body solves a 6×6 system with all lines' contributions. Each line then back-substitutes. This is the same Newton step as A, without approximation.	Keeps the current one-restraint-per-line set-up. Cost grows linearly with the number of lines, and lines can later run on different processors. A single line reproduces today's results. Needs a refactor of the coupling interface.

Static condensation, in equations

The attachment is a pin, so each line passes only a force to the body.

- **Partition.** Split line *i*'s Newton unknowns into its interior, Δu_i , and its attachment displacement, Δw_i (3 components). Its linear system is then:

$$\begin{bmatrix} K_{uu} & K_{uw} \\ K_{wu} & K_{ww} \end{bmatrix} \begin{bmatrix} \Delta u_i \\ \Delta w_i \end{bmatrix} = - \begin{bmatrix} r_u \\ r_w \end{bmatrix}$$

- **Condense.** Eliminating the interior gives the line's effective stiffness and residual at its attachment:

$$S_i = K_{ww} - K_{wu} K_{uu}^{-1} K_{uw}, \quad \tilde{r}_i = r_w - K_{wu} K_{uu}^{-1} r_u$$

Both need only the factorisation BlockEigen already computes, plus three extra back-substitutions (the columns of $K_{uu}^{-1} K_{uw}$).

- **Compatibility.** Each attachment moves with the body. With body unknowns $\Delta q = (\Delta x, \Delta \theta)$ and arm r_i (from `refAttachmentPt`):

$$\Delta w_i = T_i \Delta q, \quad T_i = [I \quad -\text{spin}(R \, r_i) \cdot T(\theta)]$$

These are the same terms as the Phase 2 rotation columns.

- **Body step.** The 6×6 body system sums every line's contribution:

$$(K_{\text{body}} + \sum_i T_i^T S_i T_i) \Delta q = - (R_{\text{body}} + \sum_i T_i^T \tilde{r}_i)$$

- **Back-substitution.** Each line recovers its interior update: $\Delta u_i = -K_{uu}^{-1} (r_u + K_{uw} T_i \Delta q)$.

K_{body} and R_{body} are the existing `rigidBodyEnd` residuals and Jacobians (inertia, gyroscopic, damping, applied load). Only the beam force term changes, from one line to a sum over lines.

Implementation steps

1. **Tidy first.** Remove the earlier staged / `fullyImplicit` path, and fix the tutorial `CQ / CM` override and the missing `abort`. Then only one coupling path needs to change.
2. **BlockEigen.** Add a way to return the condensed attachment block ($S_i, \tilde{r}_i$) from the current factorisation, and to finish the line's Newton update for a given attachment increment Δw_i . The current appended body rows become the single-line special case.
3. **beamModel interface.** Replace "evolve with the body inside" with steps a coordinator can call: assemble and condense, apply attachment increment, report convergence.
4. **Coordinator.** Move the body state and Newton loop out of a single beam into a coordinator that owns `rigidBodyEnd`; `moorFV's beamFoamCoupled` is the natural home. It collects every `finiteVolumeBeam` restraint and, at each Newton iteration:
 - a. has each line assemble and condense;
 - b. builds and solves the 6×6 system;
 - c. back-substitutes;
 - d. checks convergence of the body and every line.
5. **Diagnostics.** Add a per-line Jacobian check (finite differences of the condensed 6×6 tangent), extending `jacobianCheck`.
6. **Parallel.** Keep every line on the master process at first, as now (`Pstream::master()` in `applyRestrains` and `update`, then `Pstream::scatter` of the state). Later, lines can be spread over processors: only S_i and $\tilde{r}_i$ (12 numbers per line) need a `sumOp` reduction, and Δq a broadcast.

Validation

- **One line:** the condensed path reproduces the current monolithic results to round-off. That covers the 54 `rigidBodyEnd` checks and the `rigidBodyAndBeam_beamFoamCoupled` `interFoam` case.
 - **Standalone beamFoam with several lines:**
 - two symmetric lines, where the vertical frequency must be $\sqrt{(\Sigma k/m)}$;
 - a four-line spread mooring with an offset, compared with converged partitioned coupling;
 - a line going slack.
 - `mooringCase-H12T20` : monolithic with 1–3 outer correctors against `FvBeamNewmark` with converged correctors.
- Pass:** the 4-line case runs monolithically and matches the converged partitioned run.

Things to watch

- **Fluid added mass.** The H12T20 body floats in water, and the fluid force is still explicit. Expect to keep 2–3 outer correctors with `moveMeshOuterCorrectors yes`. What Phase 5 removes is the beam–body splitting and the need for `accelerationRelaxation`.
- **Slack lines.** S_i tends to zero when a line is slack. The body system stays well posed through its inertia term $m/(\beta\Delta t^2)$.
- **Seabed contact** (`groundContact`) only changes the line's own K_{uu} , so condensation still applies.
- **Restarts.** The coordinator must write and read the body state; `rigidBodyEnd` has no restart support yet.